

Project Report On

GYM PRO – GYM MANAGEMENT SYSTEM

Submitted in the Partial fulfillment of the requirement for awarding the degree of

Bachelor of Computer Application (BCA)

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By

Rathod Nenshi
Nandaniya Pal

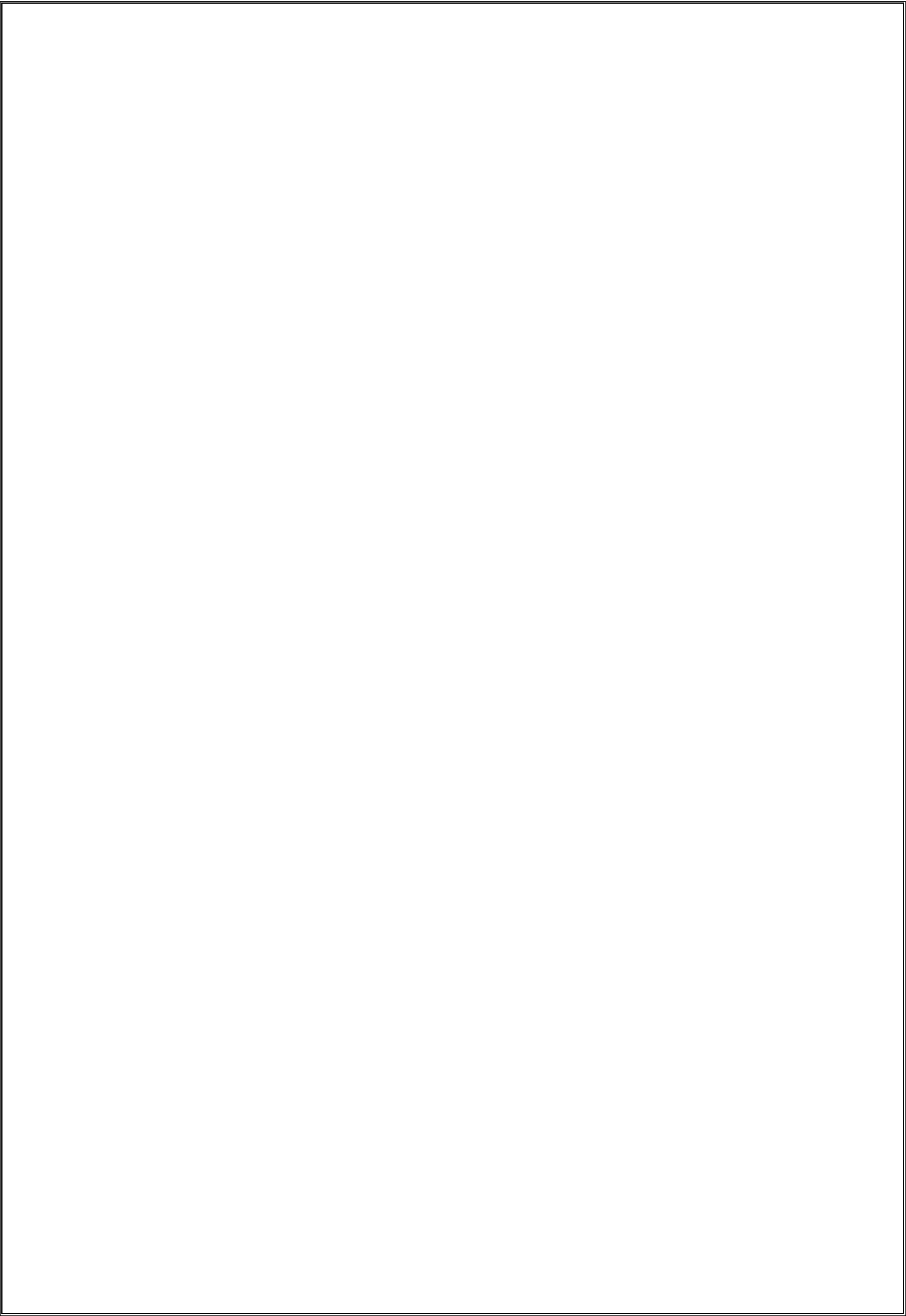
Under the Guidance of

Prof. Ishita Mem



DEPARTMENT OF COMPUTER SCIENCE
LATE P.S. HIRAPARA MAHILA BCA COLLEGE JETPUR

Affiliated to
SAURASHTRA UNIVERSITY - RAJKOT



Preface

This report contains all the outputs of a project undertaken to develop a working model of **GYM PRO – Gym Management System** and to study the existing working system. As part of this study, we analyzed the requirements of the working model and tried to improve it using the technological functionality that the current computer science provides.

This volume contains the documentation summary of the working system, the background study and analysis, the technical requirements and the working specification of the developed system.

We recommend the user to take the advantage of the study presented here and implement the system with their required improvisation. We hope this study will reduce the burden of the reanalysis of the development phase.

Acknowledgment

“It is not possible to prepare a project report without the assistance & encouragement of other people. This one is certainly no exception.”

On the very outset of this report, we would like to extend our sincere & heartfelt obligation towards all the personages who have helped us in this endeavor. Without their active guidance, help, cooperation & encouragement, we would not have made headway in the project. We are ineffably indebted to our HOD / Prof. Jitendra Kabariya for conscientious guidance and encouragement to accomplish this assignment.

We are extremely thankful and pay our gratitude to our project guide Prof. Ishita mem. her valuable guidance and support on completion of this project in it's presently.

We extend our gratitude to Department Of Computer Science, Late P.S. Hirapara Mahila BCA College-Jetpur for giving us this opportunity.

Thanking You

Student -1 Rathod Nenshi (24CS002UG05309)

Student -2 Nandaniya Pal (24CS002UG05285)

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1- Project Profile

1.1 Definition

GYM PRO – Gym Management System is a web-based software application developed to manage and maintain the daily activities of a gym. It helps the gym administrator manage member information, membership details, batches, trainers, attendance, payments and other gym-related records in a systematic and efficient way.

The system reduces manual paperwork, saves time, improves data management, and makes it easier to store, update, and retrieve gym information whenever required. It provides three separate, password-protected panels for the Admin, the Trainers and the Members, in addition to a public informational website for visitors.

1.2 Scope

The **GYM PRO – Gym Management System** is designed to manage gym members, memberships, batches, trainers, attendance and payments in an organized way. It reduces manual paperwork and makes daily gym management easier for the owner and staff.

The system helps the administrator to add, update, delete and view records easily. It saves time, reduces errors, and provides quick access to important gym information such as who is due for renewal, which batch is running short of trainers, and how much revenue has been collected.

1.3 Objective

The main objective of the **GYM PRO – Gym Management System** is to develop a simple and efficient system for managing gym activities and records. It aims to reduce manual work and make gym management faster, easier, and more organized.

The major objectives are to:

- Maintain accurate **member and membership records**.
- Manage **batches, trainers, attendance and payment information**.
- Allow a member to join more than one batch while **automatically preventing any clash** between the timings of the selected batches.
- Ensure **no membership becomes Active without a logged or verified payment**, by gating every new registration and renewal behind a payment step.
- Provide easy **adding, updating, approving, deleting and viewing** of records.
- Reduce **paperwork and manual errors**.
- Save time and improve the overall **efficiency of gym management**.

1.4 Technical Description

Technical Aspect	Description
Frontend	HTML, CSS, JavaScript
Backend	PHP
Database	MySQL

2. System Analysis and Specification

2.1 Study of Current System

Most small and medium gyms still manage their day-to-day operations manually. During the study of the existing system, the following observations were made:

- New member details, batch timings and fee amounts are written down in a membership register or note-book.
- Attendance of members is taken on a paper sign-in sheet kept at the reception desk.
- Trainers keep track of which members belong to which batch from memory or hand-written batch sheets pinned on a notice board.
- Renewal of a membership is tracked by checking a physical register or a diary maintained by the gym owner.
- Fee receipts are hand-written and a carbon copy or a simple cash-book is kept for accounts.

2.2 Weakness of Current System

The manual system currently followed by most gyms suffers from the following drawbacks:

- Registers can be lost, damaged, misplaced or duplicated, and there is no backup of the data.
- Checking whether a new admission's chosen time slot clashes with an existing batch is done manually and is error-prone; double-booking of a batch beyond its capacity is common.
- Retrieving a particular member's payment history or attendance record takes considerable time when done manually.
- There is no way for a member to check their own membership status, batch timing, attendance or dues without visiting or calling the gym.
- Sending renewal reminders or announcements to all members has to be done individually by phone or word of mouth, which is slow and unreliable.
- Trainers have no organized way to record attendance or maintain a workout plan for their batch.

2.3 Introduction of New System

GYM PRO – Gym Management System replaces the manual register with a centralized, web-based application. The new system introduces the following improvements:

- A public website lets any visitor view the gym's batches, trainers and gallery, and submit an online registration request; the Admin reviews and approves it, after which the member is created automatically.
- A member can join more than one batch at the same time, and the system automatically prevents the member from picking two batches whose timings overlap.
- Every new registration or renewal is payment-gated: the member is created with a 'Pending Payment' status and cannot reach their dashboard until a payment covering the amount due is logged by the Admin or verified online — this applies even when the Admin adds or renews a member directly.
- A member can pay online by scanning the gym's own UPI/QR code and submitting the transaction ID, which queues the claim for the Admin to verify before it is treated as a real payment; alternatively, the Admin logs the payment directly when it is collected in person.
- Registration now enforces a minimum age of 18 years, checked both on the date-picker and again on the server.
- Trainers mark attendance for their own assigned batches from their own login, and can maintain a simple workout plan for the batch.

A dashboard gives the Admin an at-a-glance view of active members, pending requests and revenue collected.

2.4 Hardware and Software Requirement (Feasibility Study)

FEASIBILITY STUDY

Feasibility study is the likelihood that whether the system will be useful to the organization or not. After studying the requirements, whether the proposed project is feasible or not is determined by checking the various feasibilities.

The main aim of the feasibility study is to determine whether developing system is financially and technically feasible. The feasibility study involves analysis of the problem and collection of data which would be input to the system, the processing required to be carried out on these data, the output data required to be produced by the system, as well as study of various constraints on the behavior of the system.

The three aspects in the feasibility study portion are:

- ✓ **Technical Feasibility**
- ✓ **Economical Feasibility**
- ✓ **Operational Feasibility**

Brief introduction of above 3 feasibilities is as under:

Technical Feasibility

Technical feasibility is considered in terms of technical requirements and their availability in the market. It determines whether the current level of technology supports the proposed system or not. The technical possibility of proposed system is as follow:

- Since the main aim of technical feasibility study is to determine whether unit possess the hardware as well as related software for the project.
- Proposed system does not require much technical detail.
- The current manual system is not so much sufficient for processing of all kinds of tasks.
- It needs low configuration computer system or high configuration computer system and the required operation system.
- It just required windows operating system.
- The organization is ready to purchase the entire required device, required by the proposed system.
- This technical specification is easily available in the market. Therefore the proposed system is technical feasible.

Technical feasibility is considered in terms of technical requirements and their availability in the market. It determines whether the current level of technology supports the proposed system or not. The technical possibility of proposed system is as follow.

In order to develop the web-application except some basic requirement there is no need of extra technology. Also no other workforce is required to develop the application. Yes, Internet is required but you can also run it off-line by sharing

Technical Feasibility

Hardware	Specification
Processor	1 GHz or Higher Intel
Motherboard	Compatible with the processor
Hard Disk	250 GB or More
RAM	1 or 2 GB
Monitor	Any CRT/LCD/LED
SMPS	450 Watt
Keyboard	Any Multimedia
Mouse	Optical

Software	Specification
Front End	HTML,PHP, CSS, JS
Back End	MySQL Server
Browser	Any web browser (Preferable Mozilla Firefox, Chrome)
Operating System	Cross Platform (Windows, Linux, etc...)

ECONOMICAL FEASIBILITY

The economical feasibility is considered in terms of money or rupees value. The organization measures the cost effectiveness of the project. The economical feasibility of the proposed system is as under.

- The organization is ready to invest in the proposed system for latest technology and better result.
- As the personnel and the manager know the computer operating, the unit need not have to appoint any computer operator.
- If any of the staff in the firm knows how to operate the computer, than the firm needs number to recruit a new person.
- The unit has to spend not much amount for the computer hardware and software.
- The organization is capable for paying the money for proposed system.
- Therefore the project possesses the economical feasibility.

The economical feasibility is considered in terms of money or rupees value. It measures the cost effectiveness of the project. The economical feasibility of the proposed system is as under.

As it is an on-line application a minimal cost of web-hosting will be charged. But it will be affordable. There would not be any charge to have look to site off-line while under construction.

Economical Feasibility

Hardware	Specification	Price
Processor	Intel core i3 10 th Gen	₹ 8200
Motherboard	GA H110	₹ 6800
Hard Disk	Seagate BarraCuda 1 TB	₹ 3400
RAM	DDR4 4GB	₹ 850
Monitor	Dell 18.5" LED	₹ 5800
SMPS	Circle 450 Watt	₹ 1200
Keyboard	Logitech Wired keyboard	₹ 750
Mouse	Logitech Optical USB	₹ 350

Software	Specification	Price
Front End	HTML, PHP, CSS, JS	Free
Back End	MySQL Server	Free
Browser	Any web browser (Preferable Mozilla Firefox)	Free
Operating System	Microsoft Windows 10 Linux (Ubuntu)	₹ 3500 Free

- Prices are approximately as per current year (2025)

Operational feasibility

Operational feasibility determines whether the proposed Gym Management System can fulfill the requirements of the gym and its users.

The system manages members, trainers, batches, attendance, payments, and memberships.

It reduces manual paperwork and makes data management easier.

Admins, trainers, and members can easily use the system with basic computer knowledge.

It provides quick and efficient access to gym information.

Therefore, the proposed Gym Management System is operationally feasible and beneficial for daily gym operations.

3. System Requirement Specification

3.1 Proposed System and Advantages

Based on the study of the current manual system and its weaknesses, the following project model (modules) is proposed for GYM PRO – Gym Management System:

1) Public Website

Provides Home, About, Batches, Gallery and Contact pages so that any visitor can learn about the gym before joining.

2) Visitor Registration

A visitor can register online by filling in personal details and choosing one or more batches. The system groups batches by time-of-day and blocks any selection whose timings overlap, before the request is sent for approval.

3) Admin Module

The Admin can manage members, trainers and batches; approve or reject registration and renewal requests; log in-person payments and verify or reject online payment submissions; post announcements; and view a dashboard of key statistics.

4) Trainer Module

A trainer can view the batches and members assigned to them, mark daily attendance, maintain a simple workout plan for the batch, and post announcements.

5) Member Module

A member can view their profile, batch(es), attendance and payment history; pay online via UPI/QR and submit the transaction ID when their status is Pending Payment; request renewal of an expiring membership; and view announcements posted by the Admin or trainer.

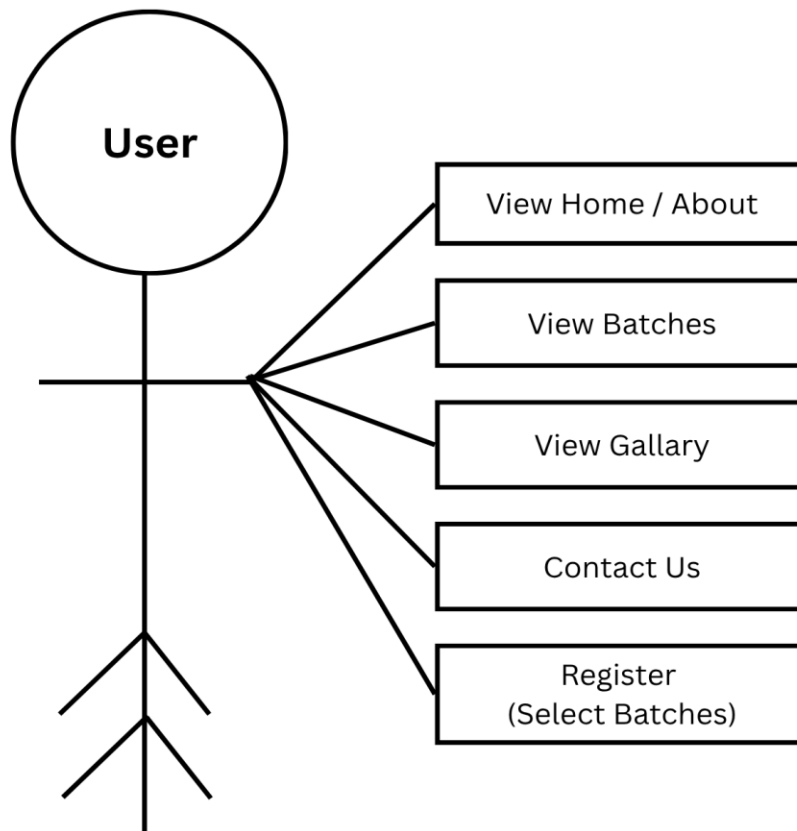
Advantages of the Proposed System

- Replaces scattered paper registers with a single, centralized, always-available database.
- Automatically prevents a member from being enrolled in two batches whose timings clash, which is not practically possible to check by hand.
- No membership can ever become Active without a logged or Admin-verified payment — the payment gate applies identically whether the member registered themselves or the Admin added them directly, closing a gap that manual systems cannot enforce.
- Lets a member pay online at their own convenience via UPI/QR, while still giving the Admin final control through a Verify/Reject step before any online claim is treated as real money received.
- Cuts down the time needed to search for a member's record, payment history or attendance from minutes to seconds.
- Gives every member and trainer self-service, read-only access to their own information, reducing the number of enquiries handled at the reception desk.
- Scales easily as the number of members, batches or trainers grows, unlike a paper register.

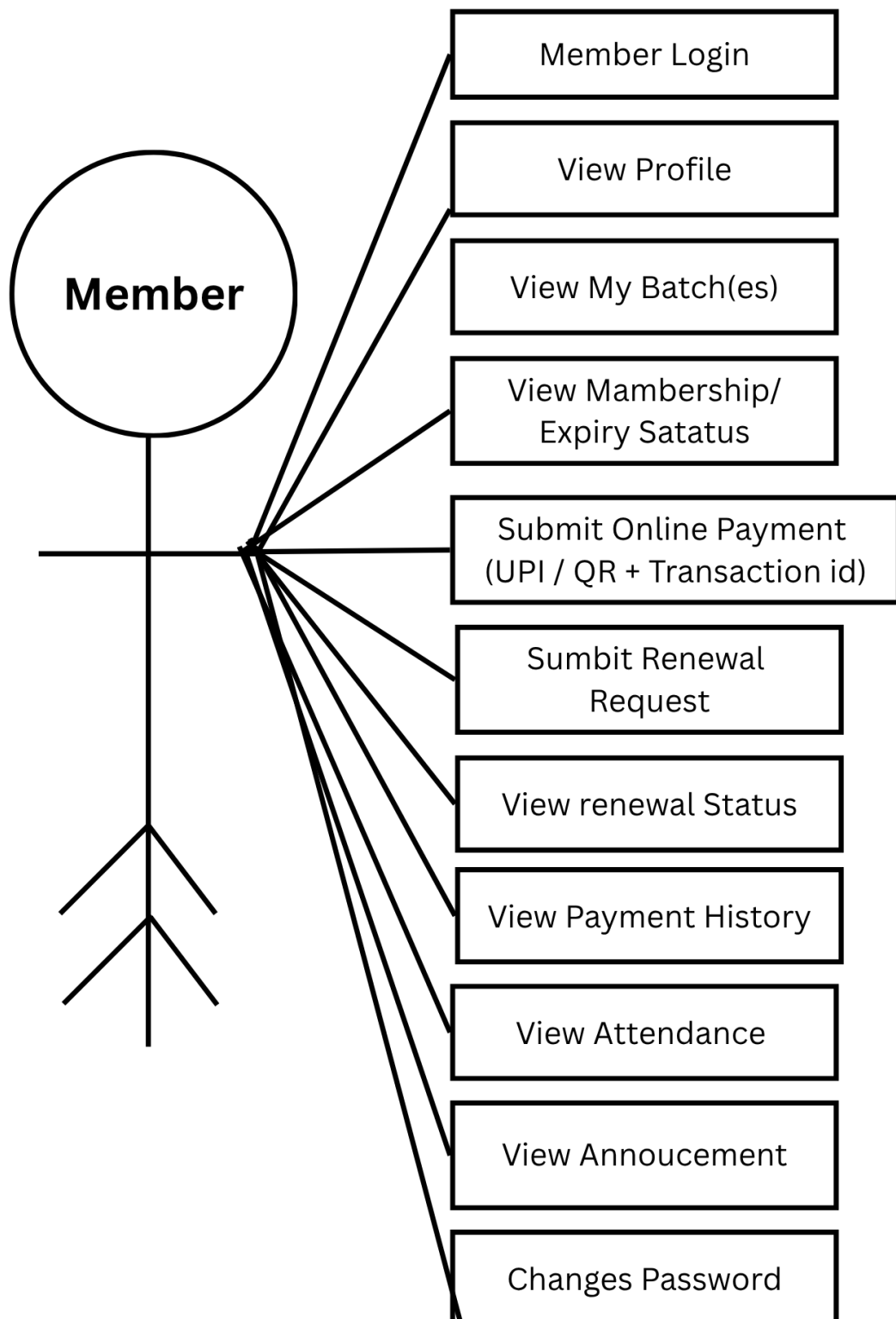
4. System Design

4.1 Use – Case Diagram

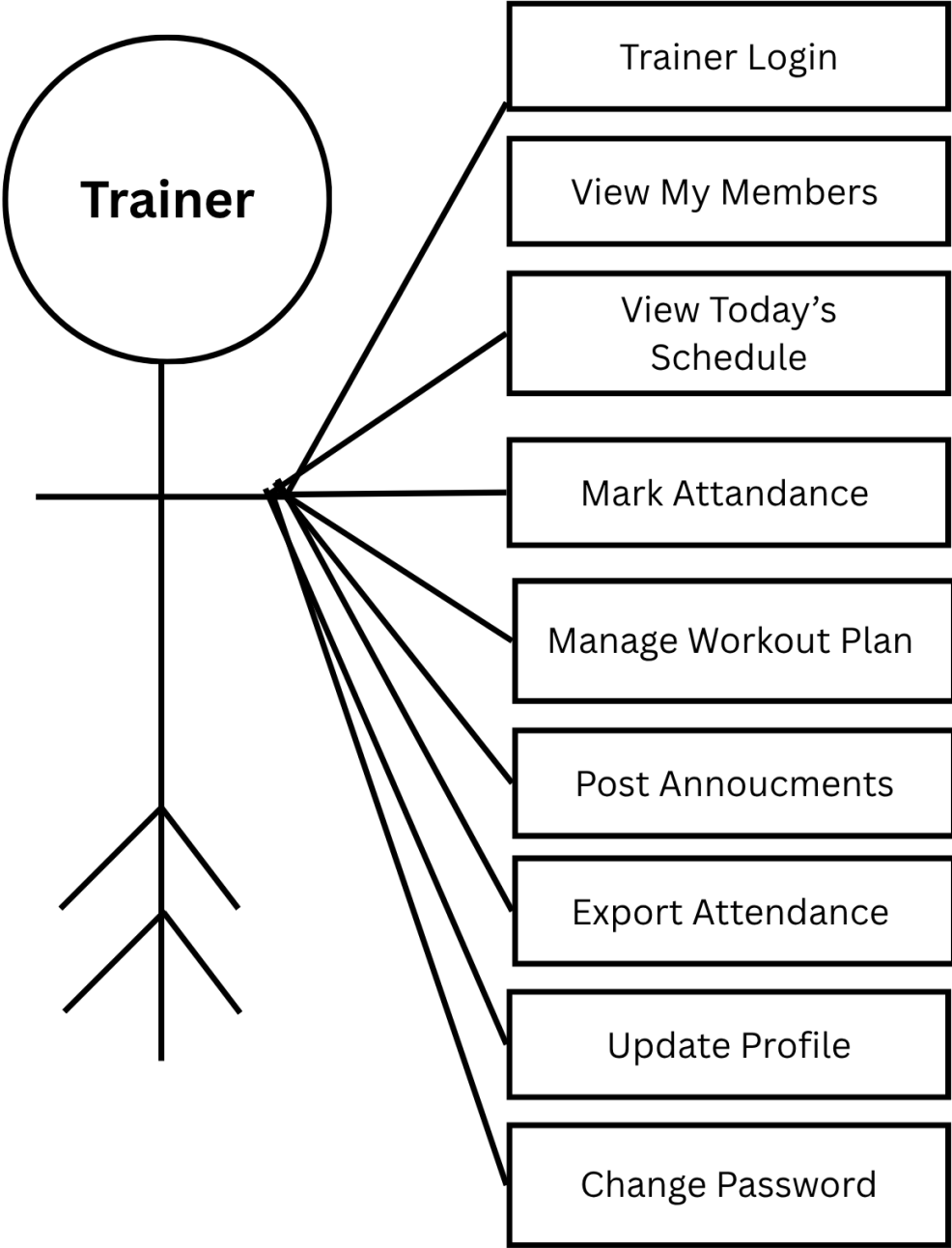
1.user



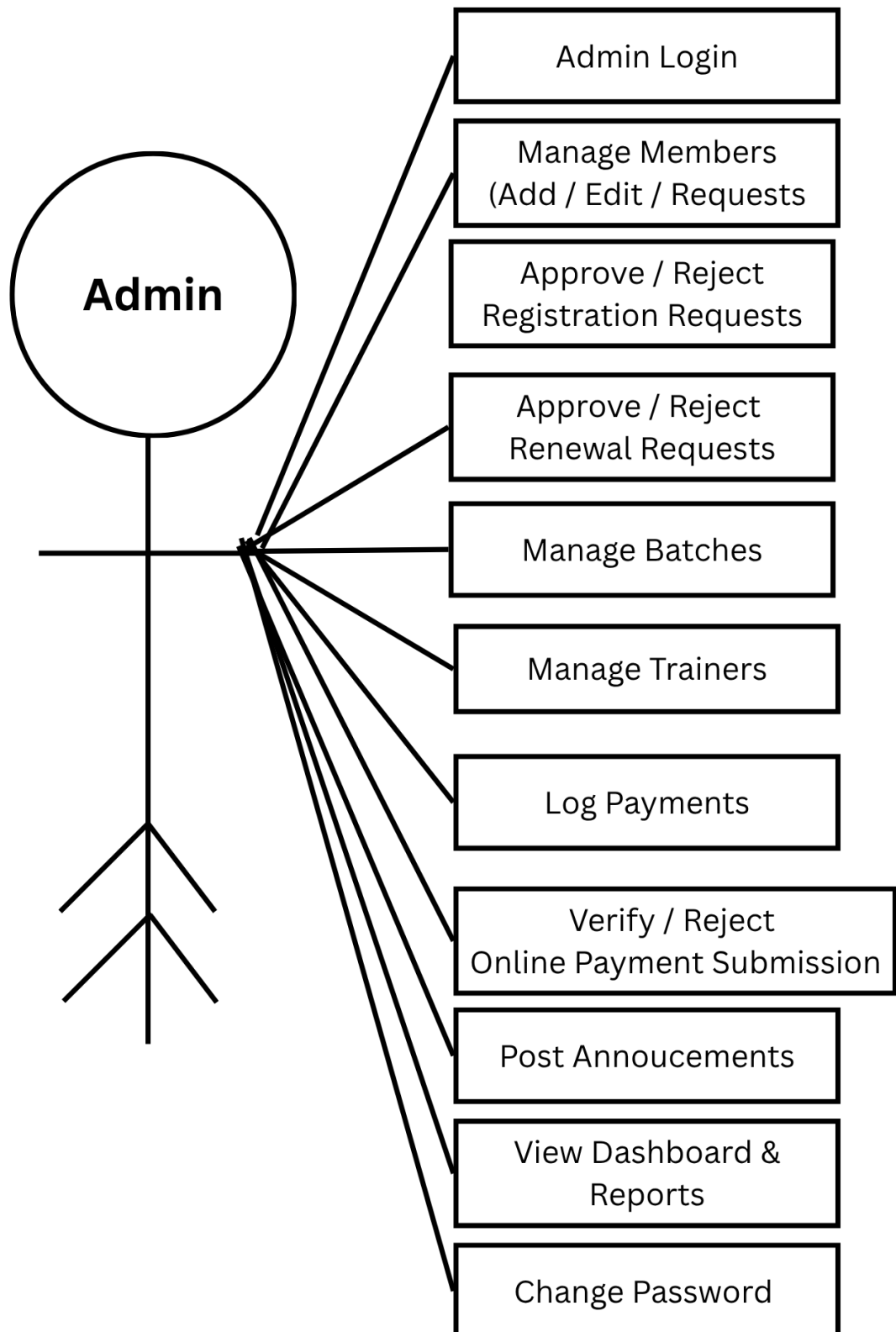
2. Member



3. Trainer

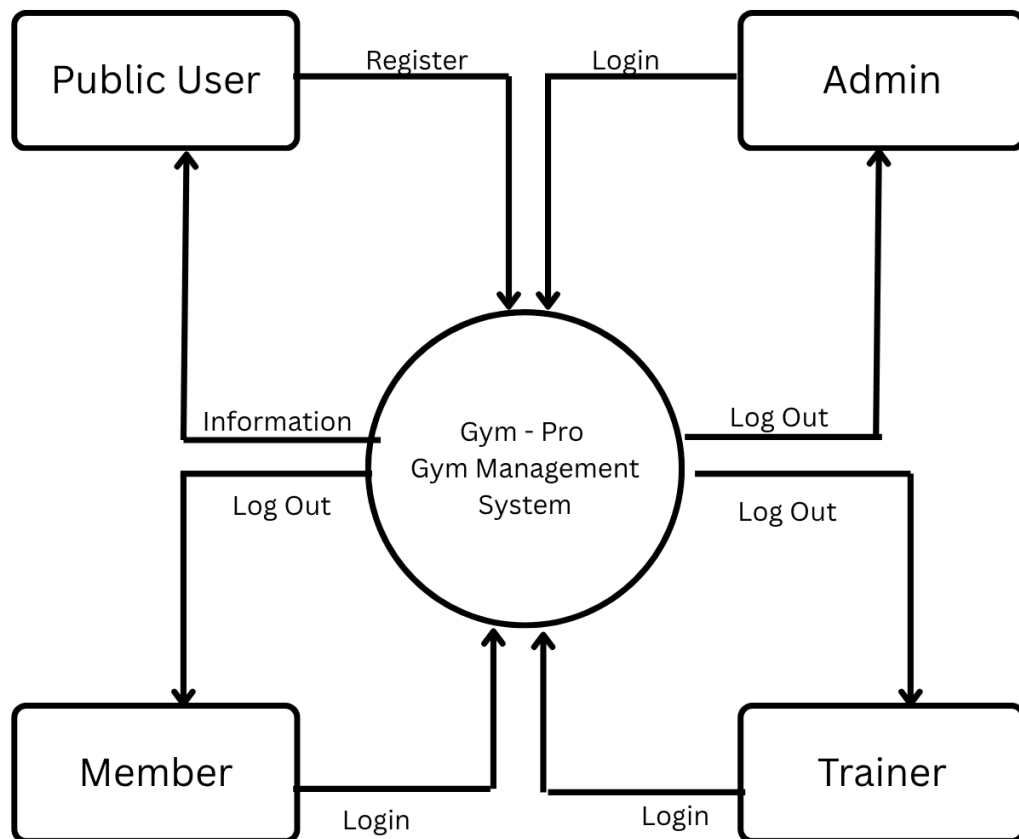


4. Admin

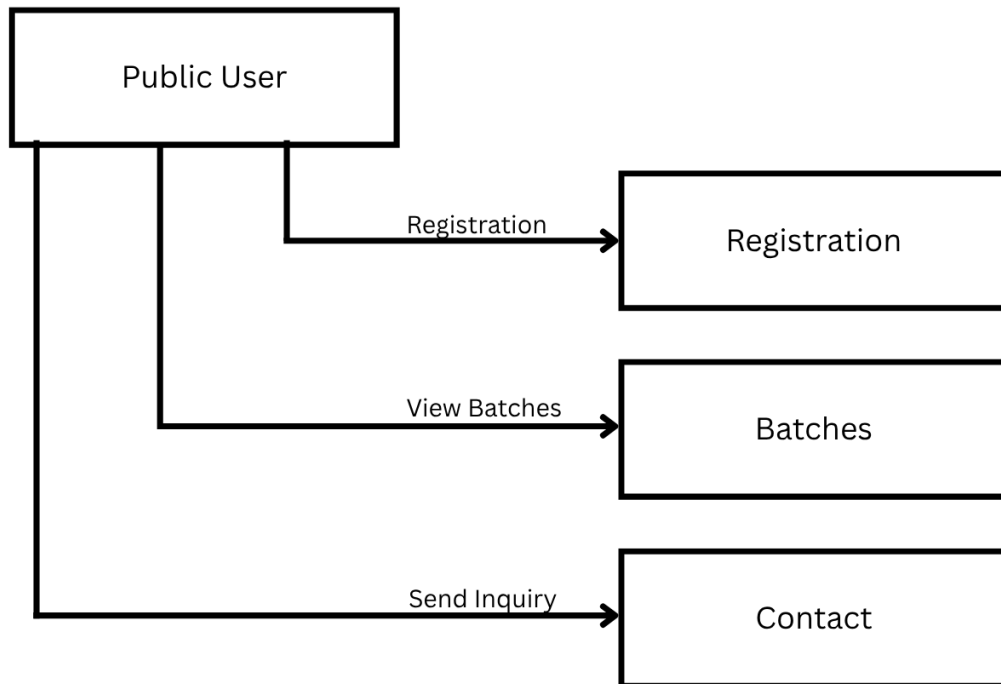


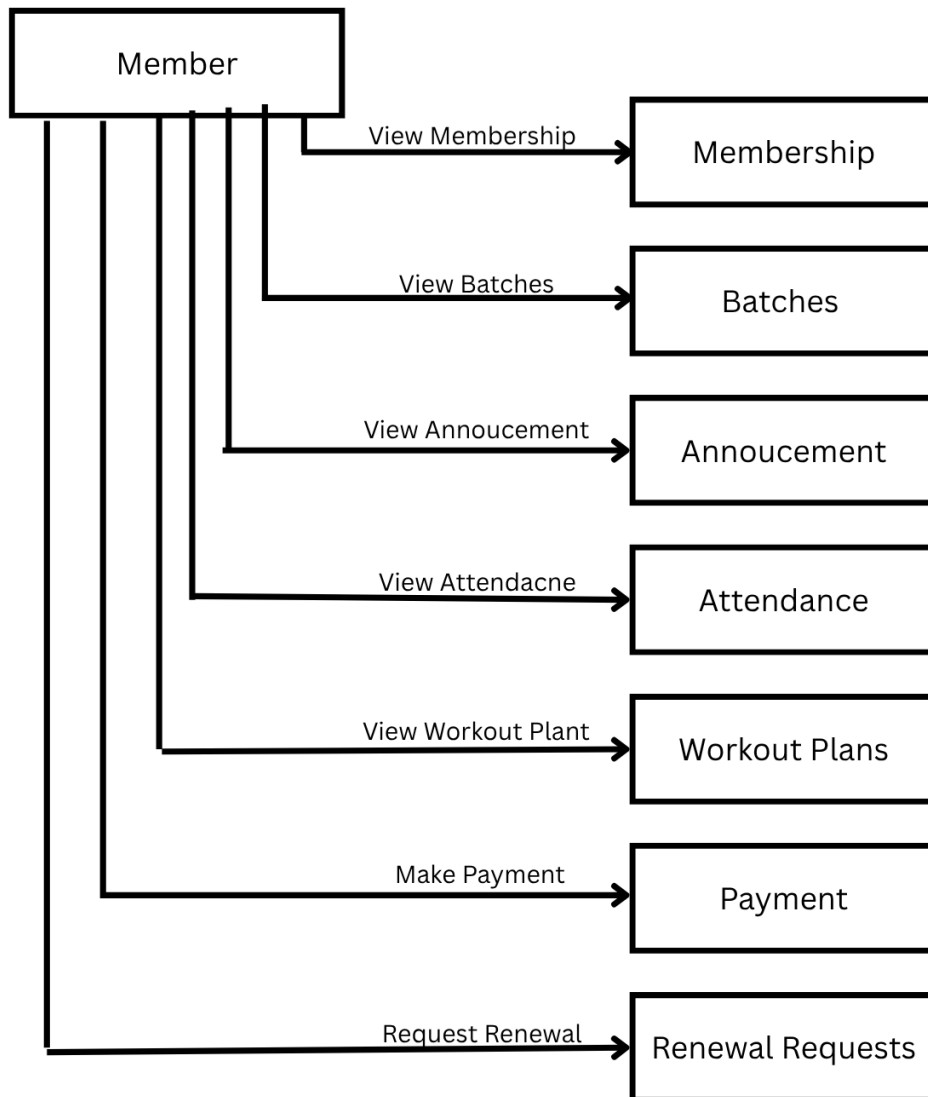
4.2 Data Flow Diagram

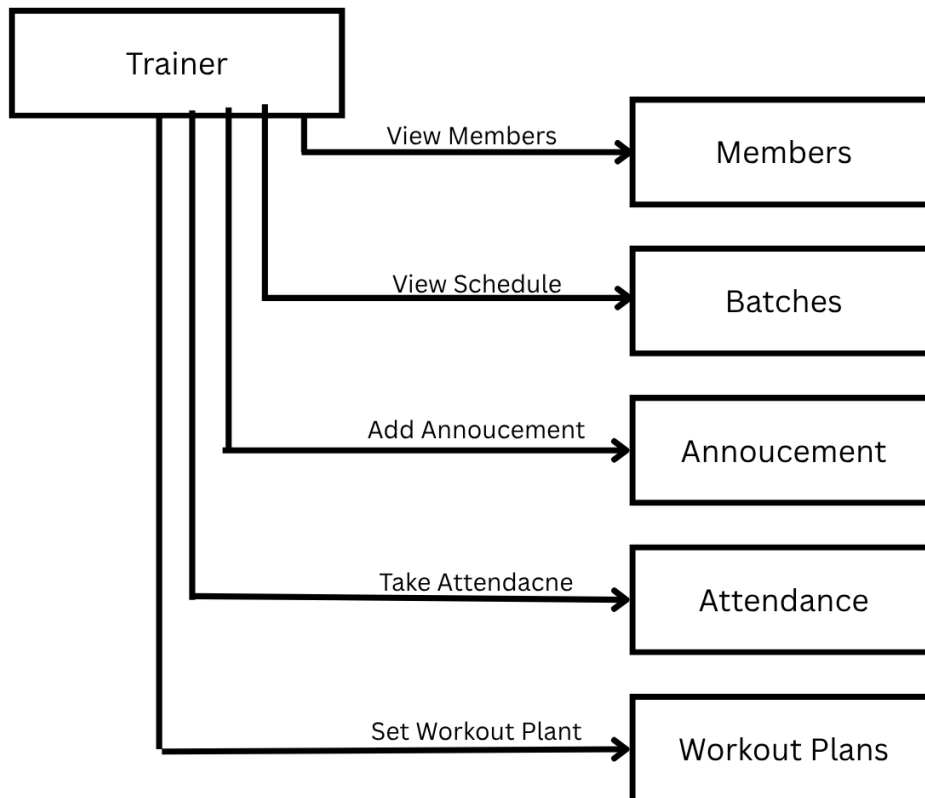
1. Level 0

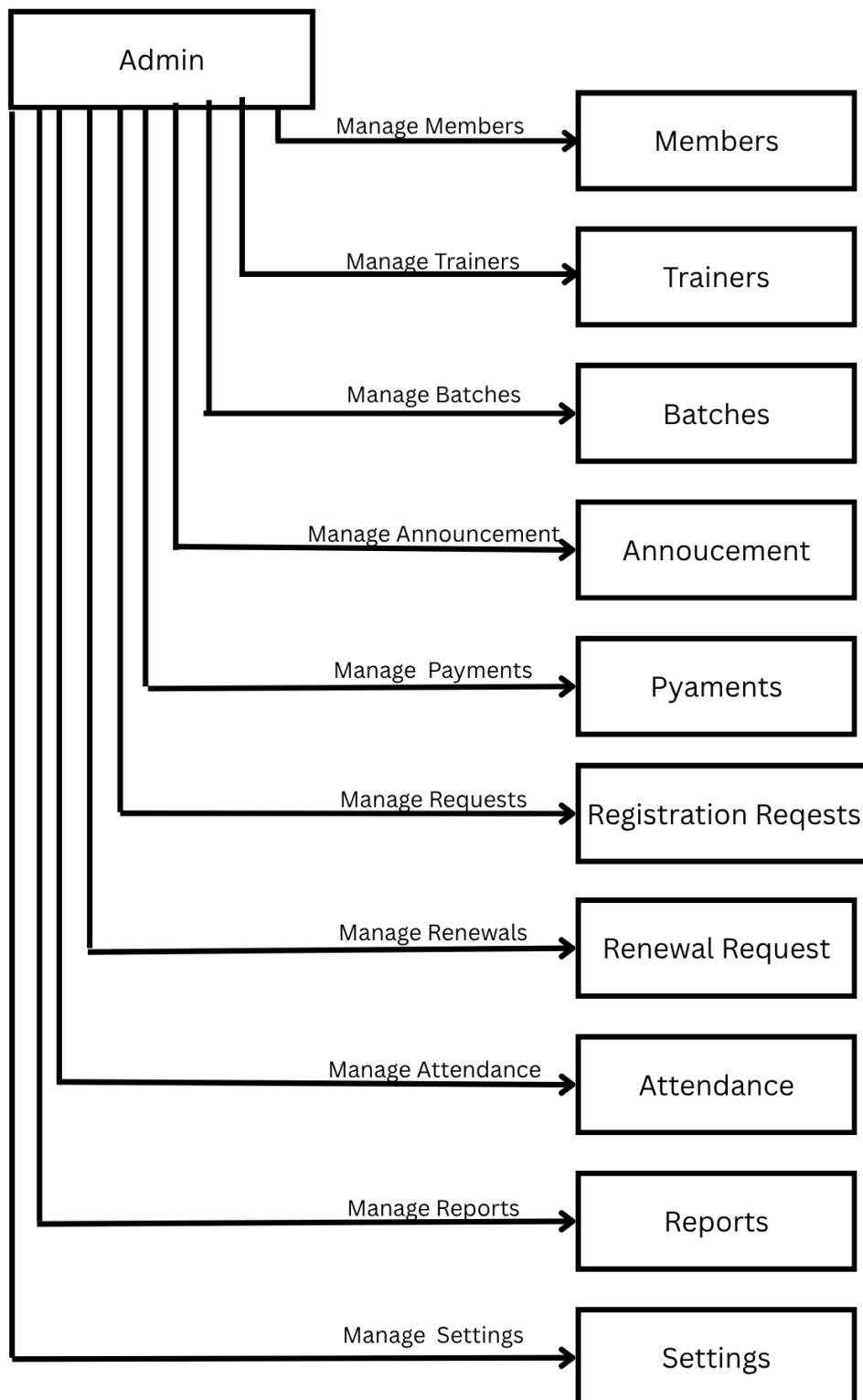


2. Level 1

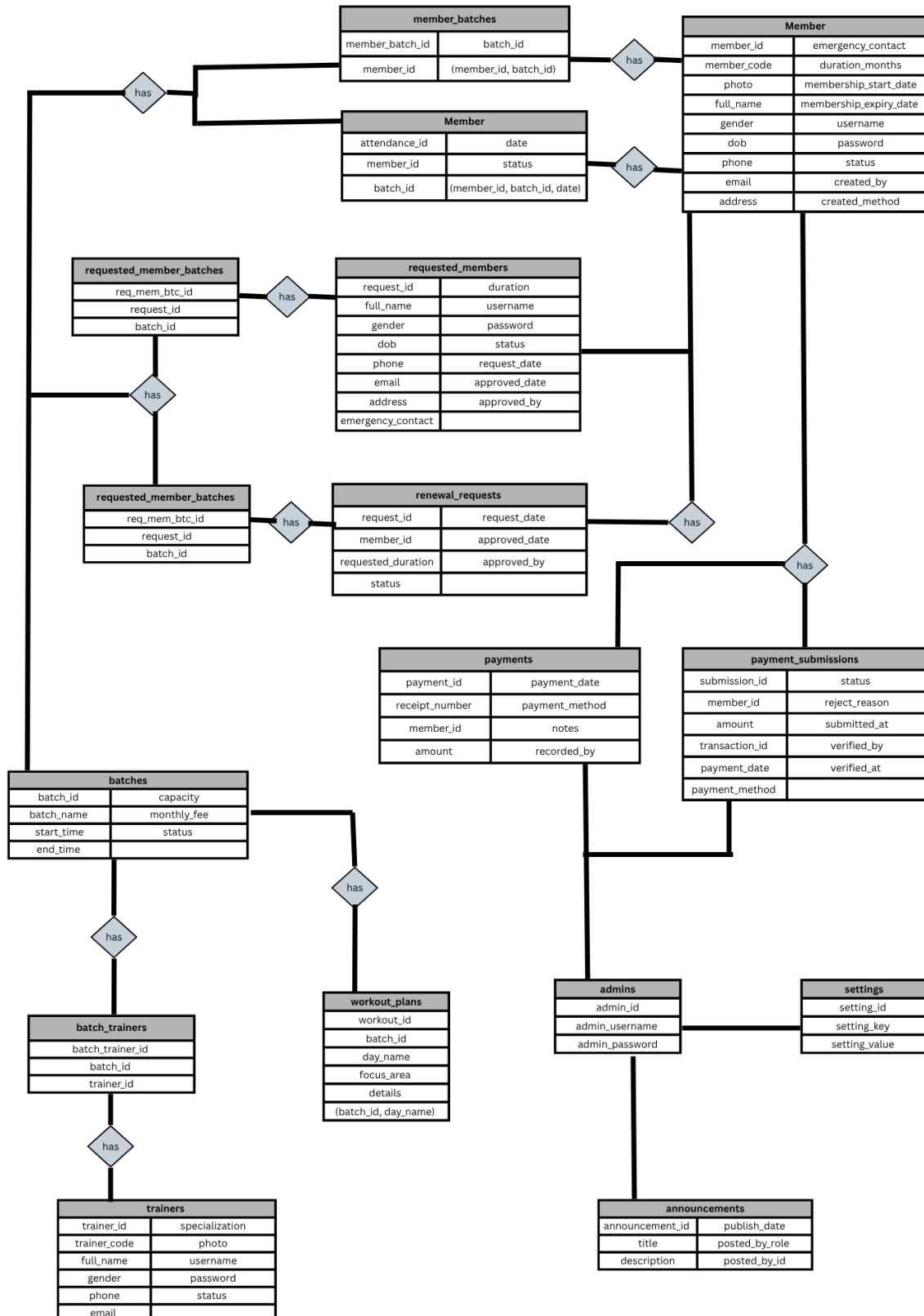








4.3 E-R Diagram



4.4 Data Dictionary

The data dictionary below lists every table of the gym_pro database exactly as created in the system (16 tables in total), showing each column, its data type and its key/constraint.

admins

Column Name	Data Type	Key / Constraint
admin_id	INT (Auto Increment)	Primary Key
admin_username	VARCHAR(50)	Not Null, Unique
admin_password	VARCHAR(255)	Not Null (hashed)

trainers

Column Name	Data Type	Key / Constraint
trainer_id	INT (Auto Increment)	Primary Key
trainer_code	VARCHAR(20)	Not Null, Unique
full_name	VARCHAR(100)	Not Null
gender	ENUM('Male','Female','Other')	Default 'Male'
phone	VARCHAR(20)	Not Null, Unique
email	VARCHAR(100)	Not Null, Unique
specialization	VARCHAR(100)	-
photo	VARCHAR(255)	Default 'default-trainer.png'
username / password	VARCHAR(50) / VARCHAR(255)	Not Null, Unique / Not Null
status	ENUM('Active','Inactive')	Default 'Active'

batches

Column Name	Data Type	Key / Constraint
batch_id	INT (Auto Increment)	Primary Key
batch_name	VARCHAR(100)	Not Null
start_time / end_time	TIME	Not Null
capacity	INT	Default 20
monthly_fee	DECIMAL(10,2)	Default 0
status	ENUM('Active','Inactive')	Default 'Active'

batch_trainers

Column Name	Data Type	Key / Constraint
batch_trainer_id	INT (Auto Increment)	Primary Key
batch_id	INT	Foreign Key -> batches
trainer_id	INT	Foreign Key -> trainers
(batch_id, trainer_id)	-	Unique Key

members

Column Name	Data Type	Key / Constraint
member_id	INT (Auto Increment)	Primary Key
member_code	VARCHAR(20)	Not Null, Unique
photo	VARCHAR(255)	Default 'default-member.png'
full_name	VARCHAR(100)	Not Null
gender	ENUM('Male','Female','Other')	Default 'Male'
dob	DATE	Not Null (18+ enforced)
phone / email	VARCHAR(20) / VARCHAR(100)	Not Null, Unique (both)
address	VARCHAR(255)	-
emergency_contact	VARCHAR(20)	-
duration_months	INT	Default 1
membership_start_date	DATE	Not Null
membership_expiry_date	DATE	Not Null
username / password	VARCHAR(50) / VARCHAR(255)	Not Null, Unique / Not Null
status	ENUM('Active','Expired','Inactive','Pending Payment')	Default 'Active'
created_by	VARCHAR(50)	-
created_method	ENUM('Request','Admin')	Default 'Admin'

member_batches

Column Name	Data Type	Key / Constraint
member_batch_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
batch_id	INT	Foreign Key -> batches
(member_id, batch_id)	-	Unique Key

payments

Column Name	Data Type	Key / Constraint
payment_id	INT (Auto Increment)	Primary Key
receipt_number	VARCHAR(20)	Not Null, Unique
member_id	INT	Foreign Key -> members
amount	DECIMAL(10,2)	Not Null
payment_date	DATE	Not Null
payment_method	ENUM('Cash','UPI','Card','Bank Transfer','Other')	Default 'Cash'
notes	VARCHAR(255)	-
recorded_by	INT	Foreign Key -> admins

payment_submissions

Column Name	Data Type	Key / Constraint
submission_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
amount	DECIMAL(10,2)	Not Null (system-calculated)
transaction_id	VARCHAR(100)	Not Null
payment_date	DATE	Not Null
payment_method	VARCHAR(20)	Default 'UPI'
status	ENUM('Pending Verification','Verified','Rejected')	Default 'Pending Verification'
reject_reason	VARCHAR(255)	-
submitted_at	DATETIME	Default Current Timestamp
verified_by	INT	Foreign Key -> admins (nullable)
verified_at	DATETIME	-

requested_members

Column Name	Data Type	Key / Constraint
request_id	INT (Auto Increment)	Primary Key
full_name, gender, dob	VARCHAR/ENUM/DATE	full_name Not Null
phone, email	VARCHAR(20) / VARCHAR(100)	Not Null
address, emergency_contact	VARCHAR	-
duration	INT	Default 1
username / password	VARCHAR(50) / VARCHAR(255)	Not Null
status	ENUM('Pending','Approved','Rejected')	Default 'Pending'
request_date / approved_date	DATETIME	Default Current / Null
approved_by	INT	Foreign Key -> admins (nullable)

requested_member_batches

Column Name	Data Type	Key / Constraint
req_mem_btc_id	INT (Auto Increment)	Primary Key
request_id	INT	Foreign Key -> requested_members
batch_id	INT	Foreign Key -> batches

renewal_requests

Column Name	Data Type	Key / Constraint
renewal_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
requested_duration	INT	Default 1
status	ENUM('Pending','Approved','Rejected')	Default 'Pending'
request_date / approved_date	DATETIME	Default Current / Null
approved_by	INT	Foreign Key -> admins (nullable)

renewal_request_batches

Column Name	Data Type	Key / Constraint
renwl_req_btc_id	INT (Auto Increment)	Primary Key
renewal_id	INT	Foreign Key -> renewal_requests
batch_id	INT	Foreign Key -> batches

announcements

Column Name	Data Type	Key / Constraint
announcement_id	INT (Auto Increment)	Primary Key
title	VARCHAR(150)	Not Null
description	TEXT	Not Null
publish_date	DATETIME	Default Current Timestamp
posted_by_role	ENUM('Admin','Trainer')	Default 'Admin'
posted_by_id	INT	Not Null

workout_plans

Column Name	Data Type	Key / Constraint
workout_id	INT (Auto Increment)	Primary Key
batch_id	INT	Foreign Key -> batches
day_name	ENUM('Monday'...'Sunday')	Not Null
focus_area	VARCHAR(100)	-
details	TEXT	-
(batch_id, day_name)	-	Unique Key

attendance

Column Name	Data Type	Key / Constraint
attendance_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
batch_id	INT	Foreign Key -> batches
date	DATE	Not Null
status	ENUM('Present','Absent')	Default 'Present'
(member_id, batch_id, date)	-	Unique Key

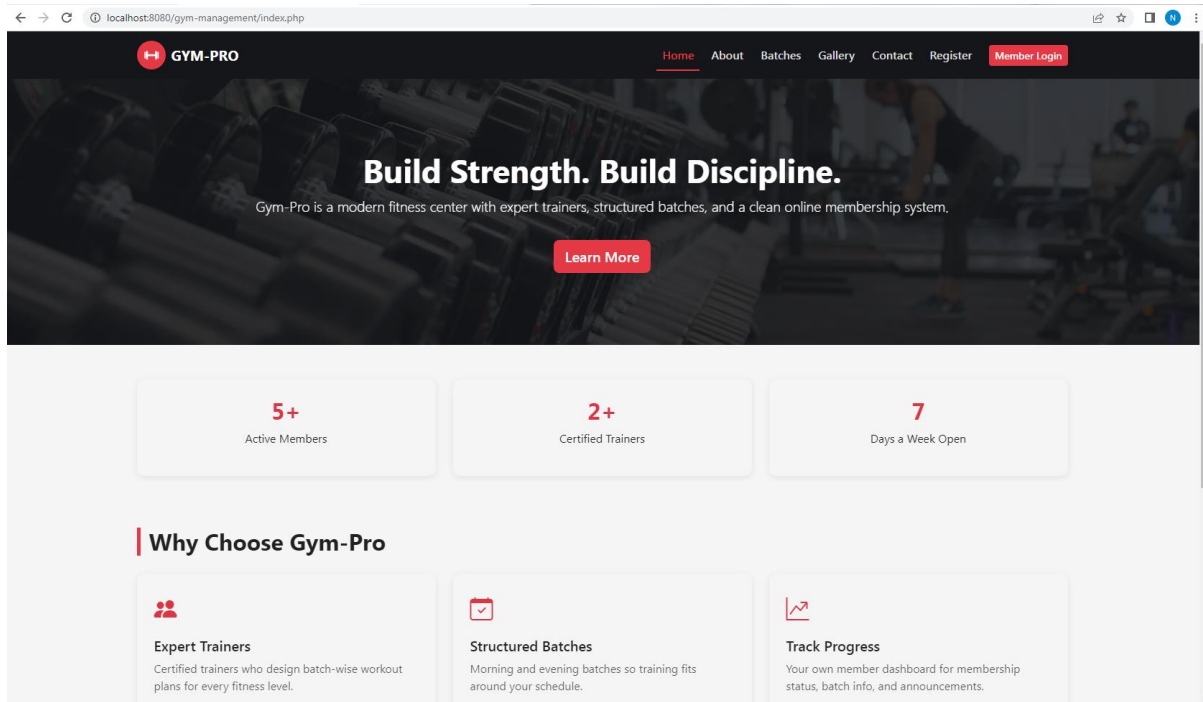
settings

Column Name	Data Type	Key / Constraint
setting_id	INT (Auto Increment)	Primary Key
setting_key	VARCHAR(50)	Not Null, Unique
setting_value	VARCHAR(255)	Not Null (e.g. gym_upi_id)

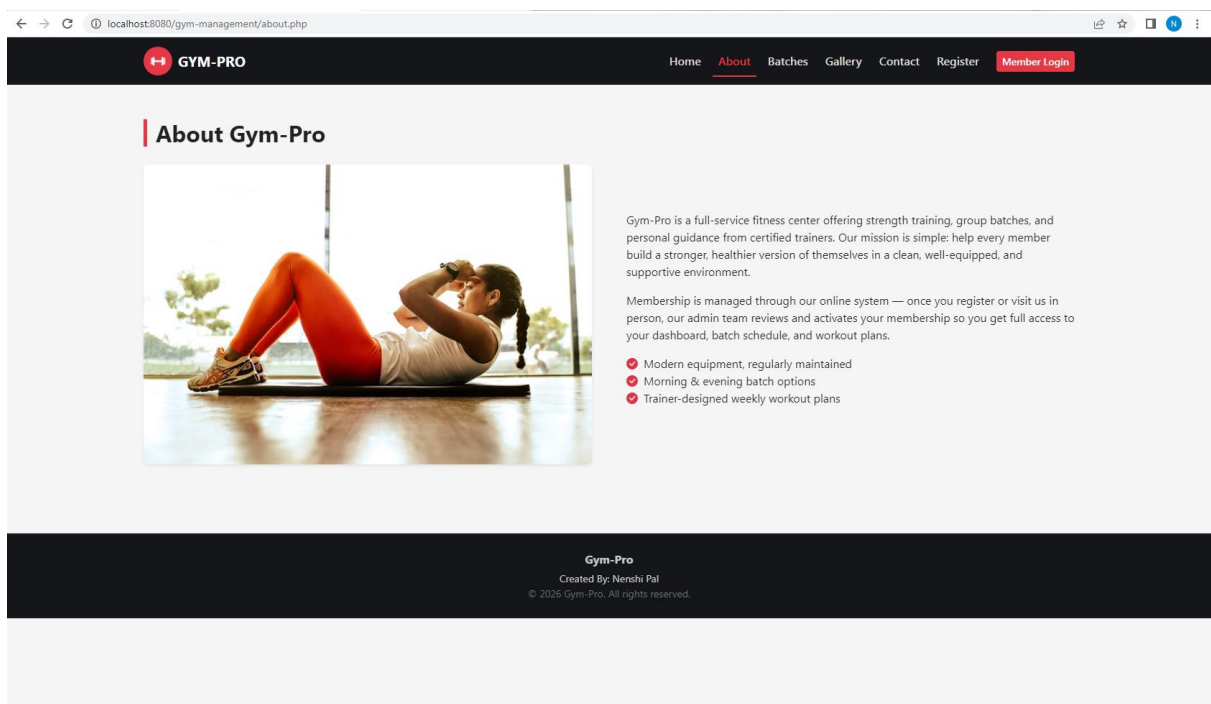
5. Screen Design

1. Public User Side

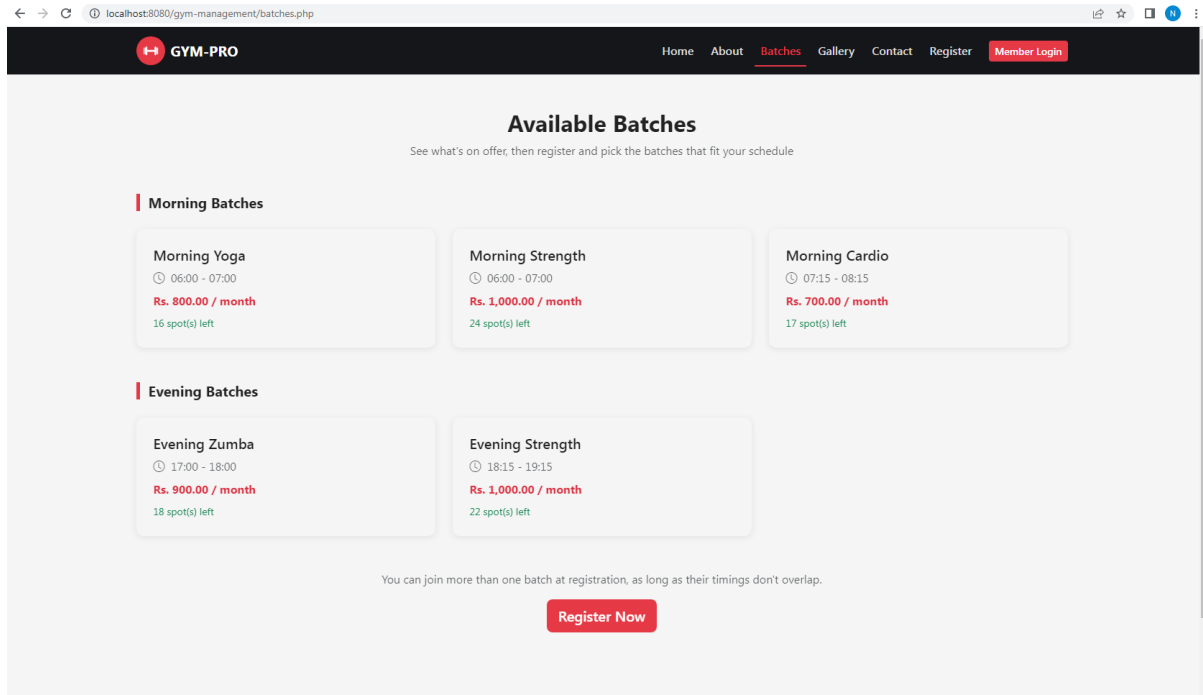
- Home



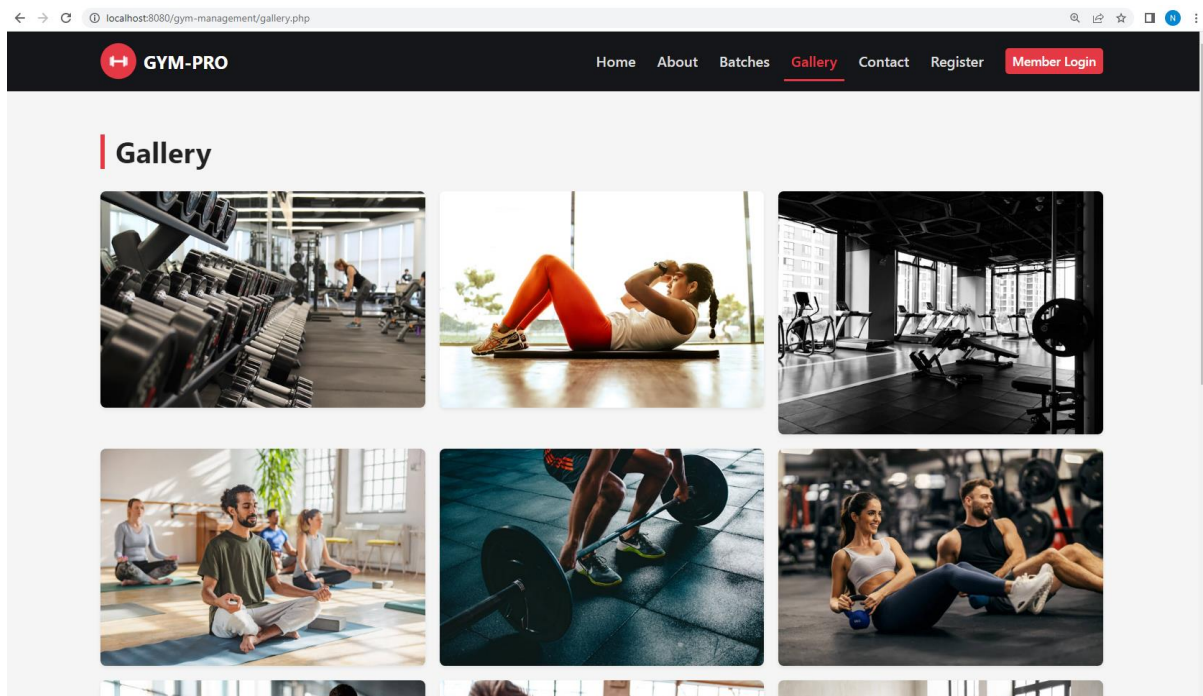
- About



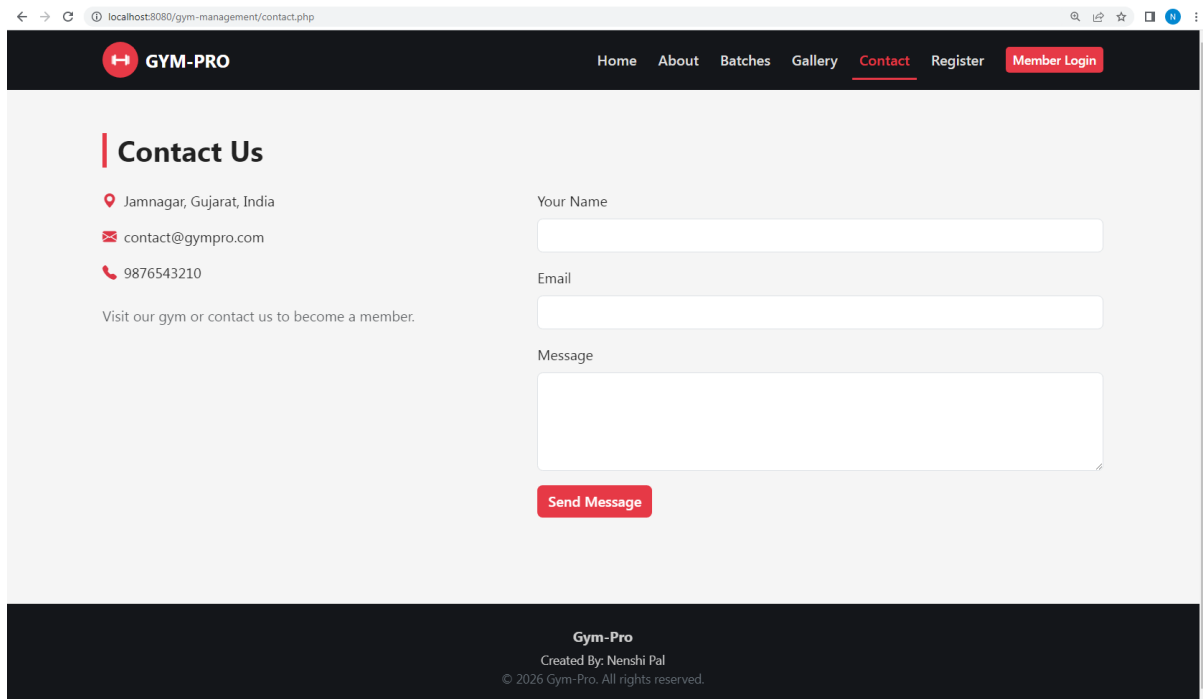
- Batches



- Gallery



- Contact



The screenshot displays a web browser window with the address bar showing 'localhost:8080/gym-management/contact.php'. The website has a dark header with the 'GYM-PRO' logo and navigation links: Home, About, Batches, Gallery, Contact (highlighted), Register, and Member Login. The main content area is titled 'Contact Us' and includes contact information: 'Jamnagar, Gujarat, India', 'contact@gympro.com', and '9876543210'. A message field with a 'Send Message' button is also present. The footer contains the text 'Gym-Pro', 'Created By: Nenshi Pal', and '© 2026 Gym-Pro. All rights reserved.'

- Register

GYM-PRO

HomeAboutBatchesGalleryContactRegisterMember Login

Membership Registration

Submit your details — an Admin will review and approve your membership.

Full Name

Gender

Date of Birth

Male

dd--

You must be at least 18 years old to register.

Phone

Email

Address

Emergency Contact

Duration (months)

1 Month

Select Batches (max 3) — choose times that don't overlap

MORNING

EVENING

☐

Morning Yoga (06:00-07:00)

Rs. 800.00 / month

☐

☐

Evening Zumba (17:00-18:00)

Rs. 900.00 / month

☐

☐

Morning Strength (06:00-07:00)

Rs. 1,000.00 / month

☐

☐

Evening Strength (18:15-19:15)

Rs. 1,000.00 / month

☐

☐

Morning Cardio (07:15-08:15)

Rs. 700.00 / month

Choose Username

Choose Password

admin

Submit Registration Request

Already a member? [Login here](#)

Gym-Pro

Created By: Nanshi Pal

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- Member Login

← → localhost:8080/gym-management/login.php

GYM-PRO Home About Batches Gallery Contact Register **Member Login**

Member Login

Username
admin

Password

Login

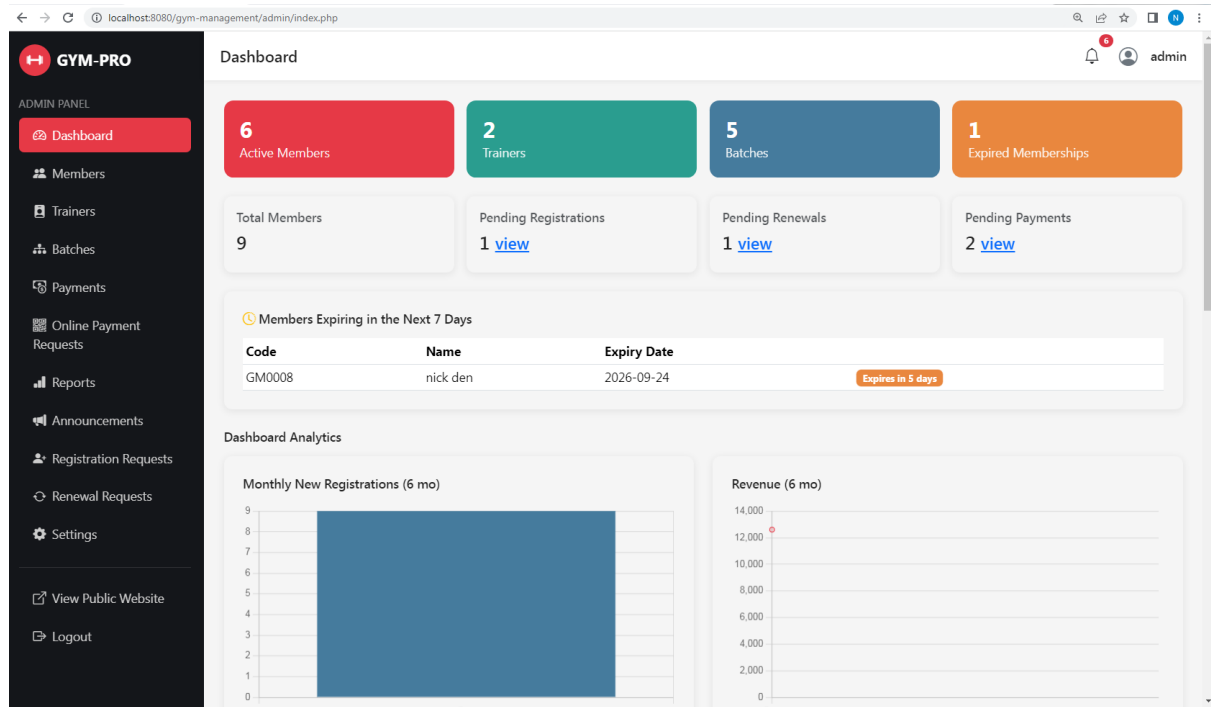
Not a member yet? [Register here](#)

[Admin / Trainer login](#)

Gym-Pro
Created By: Nenshi Pal
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2. Admin side

- Dashboard



- Members

Members

Search by name or code

[Export CSV](#) [Add Member](#)

Code	Name	Phone	Batches	Expiry Date	Status	Actions
GM0009	isha gondaliya	5555555555	—	2026-08-16	Expired	View Edit Delete
GM0008	nick den	5454545454	Morning Yoga	2026-09-24	Active	View Edit Delete
GM0007	john	2323232323	Morning Yoga, Evening Strength	2026-10-17	Pending Payment	View Edit Delete
GM0006	rathod rajveer	6351891098	Morning Yoga, Morning Cardio	2026-12-17	Active	View Edit Delete
GM0005	yashvi sarvaiya	1212121212	Morning Strength, Evening Zumba, Evening Strength	2027-03-17	Pending Payment	View Edit Delete
GM0004	vyas rachna	9000000078	Morning Yoga, Morning Cardio	2026-10-17	Active	View Edit Delete
GM0003	Kiran Joshi	9111111115	Evening Strength	2026-10-17	Active	View Edit Delete
GM0002	Sneha Patel	9111111113	Morning Cardio, Evening Zumba	2026-12-17	Active	View Edit Delete
GM0001	Amit Verma	9111111111	Morning Yoga	2026-10-17	Active	View Edit Delete

Gym-Pro | Created By: Nenshi Pal

- Trainer

GYM-PRO

ADMIN PANEL

- Dashboard
- Members
- Trainers**
- Batches
- Payments
- Online Payment Requests
- Reports
- Announcements
- Registration Requests
- Renewal Requests
- Settings
- View Public Website
- Logout

Trainers

[Add Trainer](#)

Code	Name	Phone	Specialization	Status	Actions
TR0002	Priya Nair	9000000002	Yoga & Flexibility	Active	View Edit Delete
TR0001	Rahul Sharma	9000000001	Strength & Conditioning	Active	View Edit Delete

Gym-Pro | Created By: Nenshi Pal

- Batches

GYM-PRO

ADMIN PANEL

- Dashboard
- Members
- Trainers
- Batches**
- Payments
- Online Payment Requests
- Reports
- Announcements
- Registration Requests
- Renewal Requests
- Settings
- View Public Website
- Logout

Batches

[Add Batch](#)

Batch Name	Time	Capacity	Filled	Fee/mo	Status	Actions
Evening Strength	18:15 - 19:15	25	Rs. 1,000.00	3	Active	View Edit Delete
Evening Zumba	17:00 - 18:00	20	Rs. 900.00	2	Active	View Edit Delete
Morning Cardio	07:15 - 08:15	20	Rs. 700.00	3	Active	View Edit Delete
Morning Strength	06:00 - 07:00	25	Rs. 1,000.00	1	Active	View Edit Delete
Morning Yoga	06:00 - 07:00	20	Rs. 800.00	5	Active	View Edit Delete

Gym-Pro | Created By: Nenshi Pal

- Payment

Payments

Search by member name, code, or receipt

Pending Payments Export CSV Log Payment

Receipt #	Member	Amount	Date	Method	Notes	Actions
RC0006	rathod rajveer (GM0006)	Rs. 4,500.00	2026-09-17	Cash		Receipt
RC0005	Sneha Patel (GM0002)	Rs. 1,800.00	2026-09-17	Cash		Receipt
RC0003	Kiran Joshi (GM0003)	Rs. 1,000.00	2026-09-17	UPI	Verified online payment — Transaction ID: 123456789012	Receipt
RC0002	Sneha Patel (GM0002)	Rs. 3,000.00	2026-09-17	UPI	Partial payment for 3-month renewal	Receipt
RC0001	Amit Verma (GM0001)	Rs. 800.00	2026-09-17	Cash	First month payment	Receipt
RC0004	vyas rachna (GM0004)	Rs. 1,500.00	2026-09-01	UPI	Verified online payment — Transaction ID: 4545454545	Receipt

Gym-Pro | Created By: Nenshi Pal

- Online Payment Request

Online Payment Requests

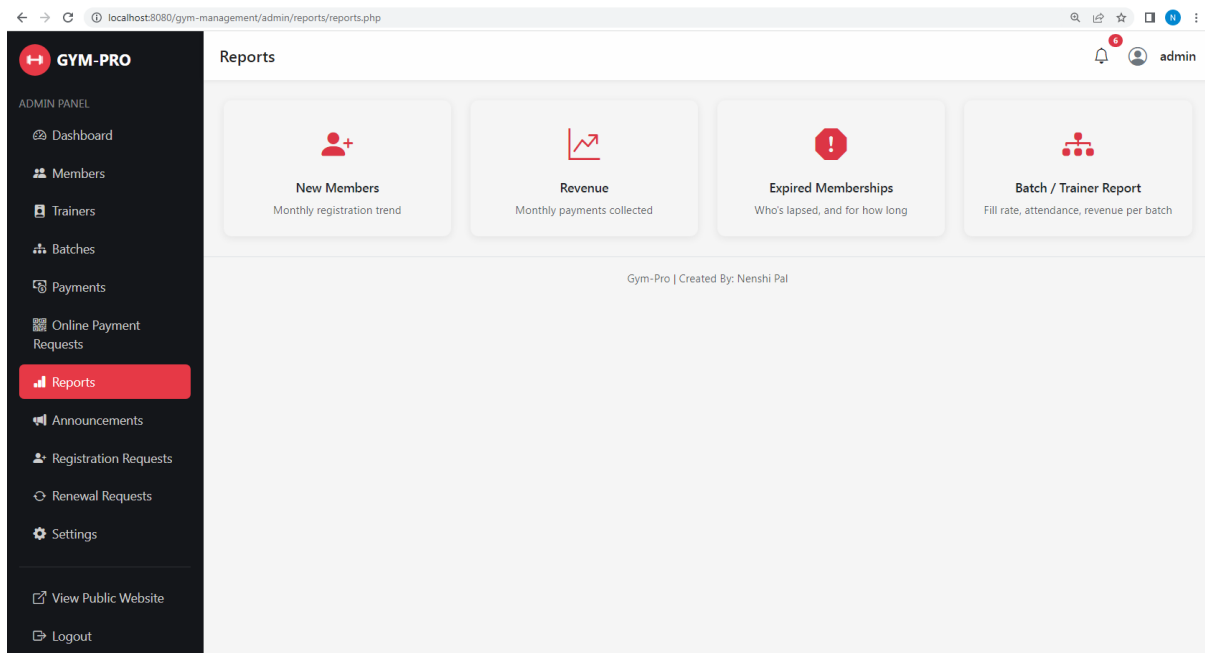
Pending Verification Verified Rejected All

Back to Payment History

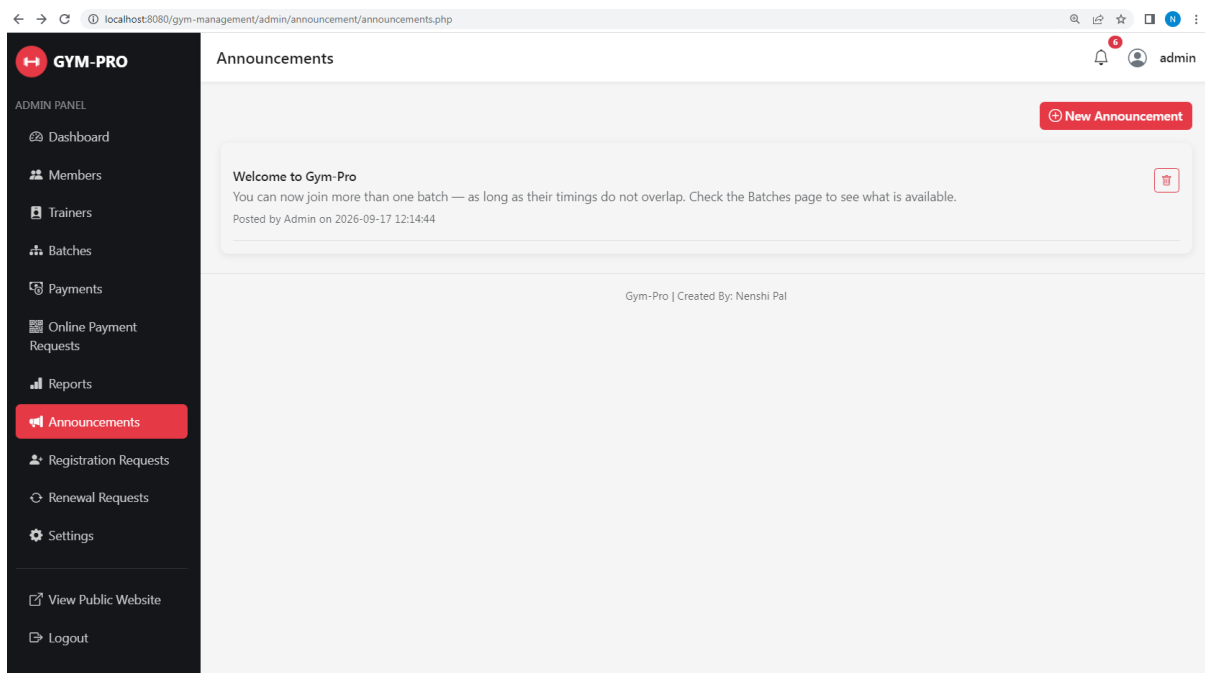
Member	Amount	Transaction ID	Payment Date	Submitted	Status	Actions
john (GM0007)	Rs. 1,800.00	4545454545	2026-09-19	2026-09-19 09:17:17	Pending Verification	View
yashvi sarvaiya (GM0005)	Rs. 17,400.00	7878787hghj	2026-09-17	2026-09-17 12:39:35	Pending Verification	View

Gym-Pro | Created By: Nenshi Pal

- Reports



- Announcement



- Registration requests

The screenshot shows the GYM-PRO admin interface. The sidebar on the left contains the following links: ADMIN PANEL, Dashboard, Members, Trainers, Batches, Payments, Online Payment Requests, Reports, Announcements, Registration Requests (highlighted), Renewal Requests, Settings, View Public Website, and Logout. The main content area is titled 'Registration Requests' and features a filter bar with 'Pending', 'Approved', 'Rejected', and 'All' tabs. Below the filter is a table with the following data:

Name	Phone	Requested Batches	Requested On	Status	Actions
krish gujrati	9824923299	Morning Strength, Morning Cardio, Evening Strength	2026-09-19 10:15:52	Pending	View Approve Reject

At the bottom of the main content area, it says 'Gym-Pro | Created By: Nenshi Pal'.

- Renewal requests

The screenshot shows the GYM-PRO admin interface. The sidebar on the left contains the following links: ADMIN PANEL, Dashboard, Members, Trainers, Batches, Payments, Online Payment Requests, Reports, Announcements, Registration Requests, Renewal Requests (highlighted), Settings, View Public Website, and Logout. The main content area is titled 'Renewal Requests' and features a filter bar with 'Pending', 'Approved', 'Rejected', and 'All' tabs. Below the filter is a table with the following data:

Member	Requested Batches	Duration	Requested On	Status	Actions
isha gondaliya (GM0009)	Morning Yoga, Morning Cardio	1 mo	2026-09-19 10:18:17	Pending	View Approve Reject

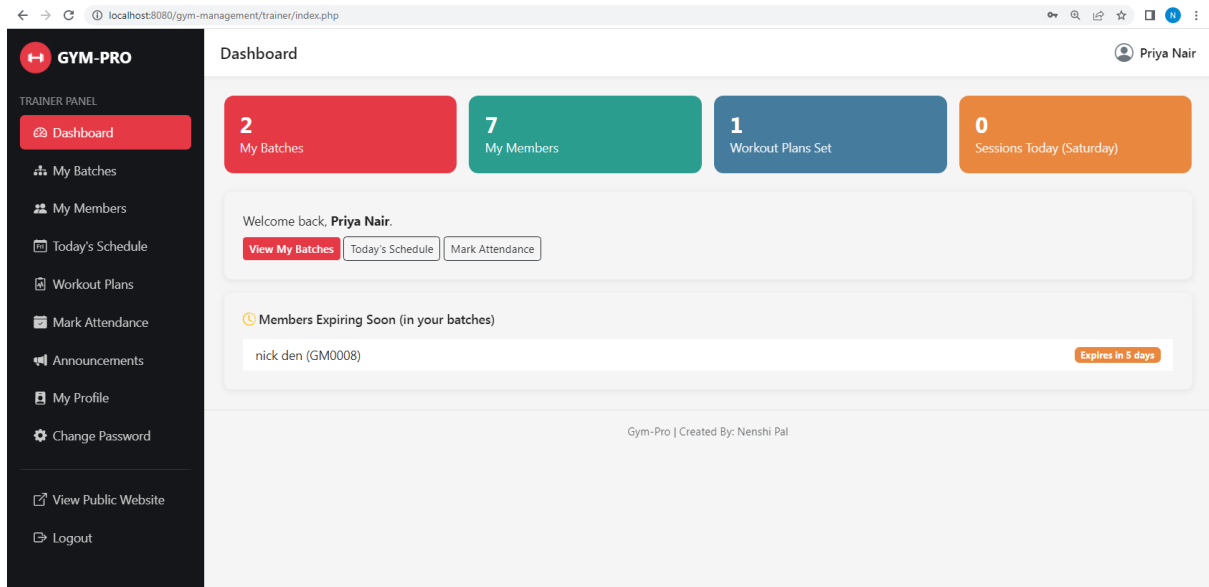
At the bottom of the main content area, it says 'Gym-Pro | Created By: Nenshi Pal'.

- Setting

The screenshot displays the 'Settings' page of the 'GYM-PRO' admin panel. The browser's address bar shows the URL 'localhost:8080/gym-management/admin/settings/settings.php'. The left sidebar, titled 'ADMIN PANEL', lists various navigation options: Dashboard, Members, Trainers, Batches, Payments, Online Payment Requests, Reports, Announcements, Registration Requests, Renewal Requests, Settings (highlighted in red), View Public Website, and Logout. The main content area is titled 'Settings' and features a 'Change Password' form. This form includes three input fields: 'Current Password', 'New Password', and 'Confirm New Password', each with a toggle icon for password visibility. A red 'Update Password' button is positioned below the 'Confirm New Password' field. At the bottom of the main content area, a footer reads 'Gym-Pro | Created By: Nenshi Pal'. The top right corner of the page shows a notification bell icon and a user profile icon labeled 'admin'.

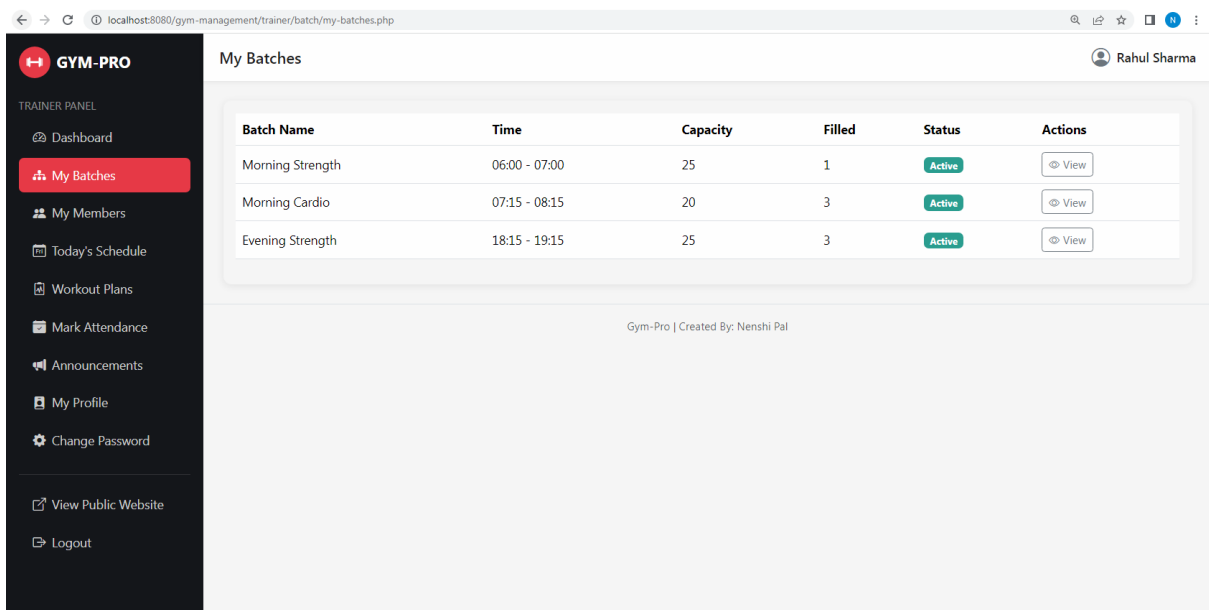
3. Trainer side

- Dashboard



The screenshot shows the 'Dashboard' page for a trainer named Priya Nair. The page features a dark sidebar with navigation links: Dashboard, My Batches, My Members, Today's Schedule, Workout Plans, Mark Attendance, Announcements, My Profile, Change Password, View Public Website, and Logout. The main content area displays four summary cards: 2 My Batches, 7 My Members, 1 Workout Plans Set, and 0 Sessions Today (Saturday). Below these cards, a welcome message is followed by buttons for 'View My Batches', 'Today's Schedule', and 'Mark Attendance'. A section titled 'Members Expiring Soon (in your batches)' lists a member 'nick den (GM0008)' with an 'Expires in 5 days' warning. The footer indicates 'Gym-Pro | Created By: Nenshi Pal'.

- My Batches

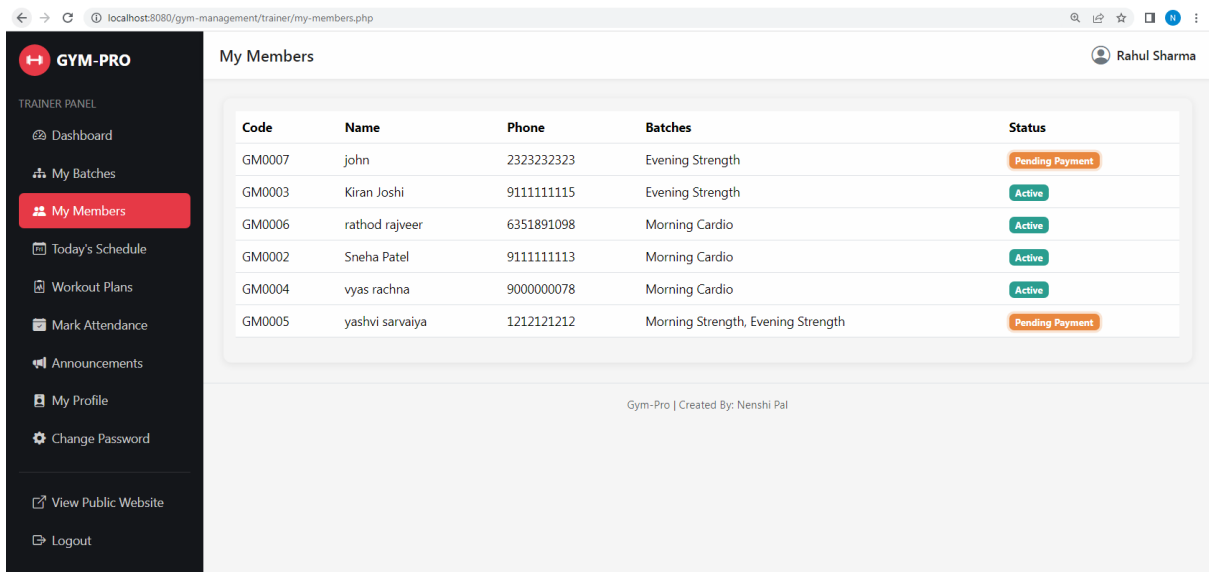


The screenshot shows the 'My Batches' page for a trainer named Rahul Sharma. The page features a dark sidebar with navigation links: Dashboard, My Batches, My Members, Today's Schedule, Workout Plans, Mark Attendance, Announcements, My Profile, Change Password, View Public Website, and Logout. The main content area displays a table with the following data:

Batch Name	Time	Capacity	Filled	Status	Actions
Morning Strength	06:00 - 07:00	25	1	Active	View
Morning Cardio	07:15 - 08:15	20	3	Active	View
Evening Strength	18:15 - 19:15	25	3	Active	View

The footer indicates 'Gym-Pro | Created By: Nenshi Pal'.

- My Members

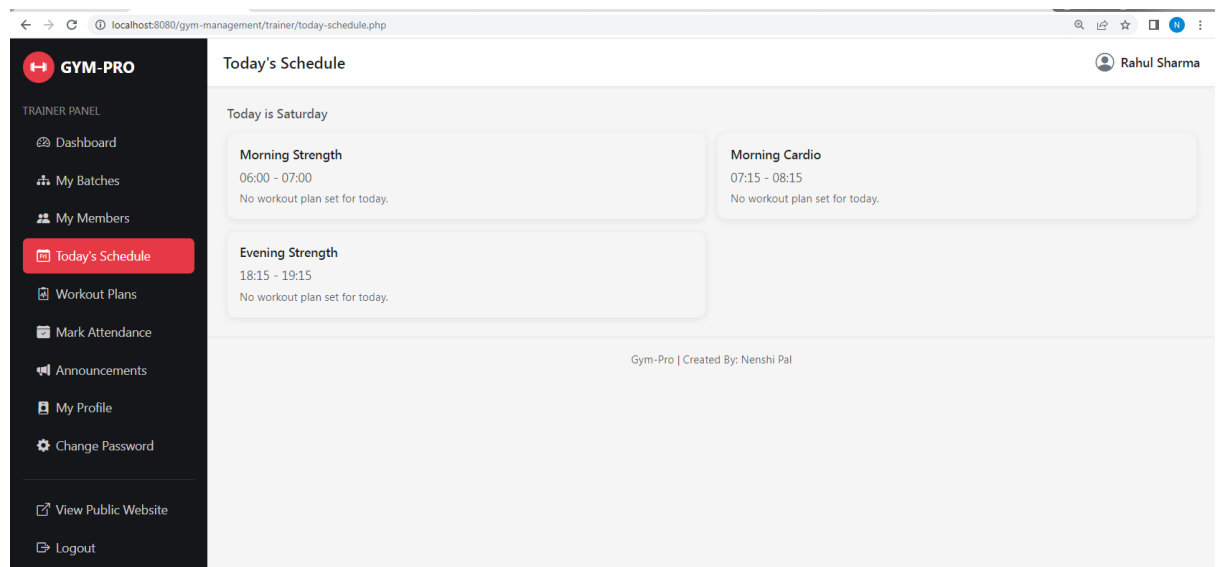


The screenshot displays the 'My Members' page of the GYM-PRO application. The left sidebar contains a 'TRAINER PANEL' with options: Dashboard, My Batches, My Members (highlighted), Today's Schedule, Workout Plans, Mark Attendance, Announcements, My Profile, Change Password, View Public Website, and Logout. The main content area shows a table of members:

Code	Name	Phone	Batches	Status
GM0007	john	2323232323	Evening Strength	Pending Payment
GM0003	Kiran Joshi	9111111115	Evening Strength	Active
GM0006	rathod rajveer	6351891098	Morning Cardio	Active
GM0002	Sneha Patel	9111111113	Morning Cardio	Active
GM0004	vyas rachna	9000000078	Morning Cardio	Active
GM0005	yashvi sarvalya	1212121212	Morning Strength, Evening Strength	Pending Payment

At the bottom of the page, it says 'Gym-Pro | Created By: Nenshi Pal'.

- Today's Schedule



The screenshot displays the 'Today's Schedule' page of the GYM-PRO application. The left sidebar is the same as the previous page, with 'Today's Schedule' highlighted. The main content area shows the schedule for Saturday:

Today is Saturday

Morning Strength
06:00 - 07:00
No workout plan set for today.

Morning Cardio
07:15 - 08:15
No workout plan set for today.

Evening Strength
18:15 - 19:15
No workout plan set for today.

At the bottom of the page, it says 'Gym-Pro | Created By: Nenshi Pal'.

- Workout Plans

GYM-PRO

TRAINER PANEL

- Dashboard
- My Batches
- My Members
- Today's Schedule
- Workout Plans**
- Mark Attendance
- Announcements
- My Profile
- Change Password
- View Public Website
- Logout

Workout Plans

Choose Batch

Morning Cardio

Monday

Focus area (e.g. Chest & Triceps)

Details (exercises, sets, reps)

Tuesday

Focus area (e.g. Chest & Triceps)

Details (exercises, sets, reps)

Wednesday

Focus area (e.g. Chest & Triceps)

Details (exercises, sets, reps)

Thursday

Focus area (e.g. Chest & Triceps)

Details (exercises, sets, reps)

Friday

Focus area (e.g. Chest & Triceps)

Details (exercises, sets, reps)

Saturday

Focus area (e.g. Chest & Triceps)

Details (exercises, sets, reps)

Sunday

Focus area (e.g. Chest & Triceps)

Details (exercises, sets, reps)

Save Workout Plan

- Mark Attendance

GYM-PRO

TRAINER PANEL

- Dashboard
- My Batches
- My Members
- Today's Schedule
- Workout Plans
- Mark Attendance**
- Announcements
- My Profile
- Change Password
- View Public Website
- Logout

Mark Attendance

Batch

Morning Cardio

Date

19-Sep-2026

Load

Export Attendance Sheet (CSV)

Code	Name	Status
GM0006	rathod rajveer	Present Absent
GM0002	Sneha Patel	Present Absent
GM0004	vyas rachna	Present Absent

Save Attendance

Gym-Pro | Created By: Nenshi Pal

- My Profile

← → ↻ localhost:8080/gym-management/trainer/profile.php 🔍 🖨 ⭐ 📱 🌙

GYM-PRO

TRAINER PANEL

- Dashboard
- My Batches
- My Members
- Today's Schedule
- Workout Plans
- Mark Attendance
- Announcements
- My Profile**
- Change Password
- View Public Website
- Logout

My Profile

Rahul Sharma (TR0001)

Gender	Male
Phone	9000000001
Email	rahul.trainer@gympro.com
Specialization	Strength & Conditioning
Status	Active

To change contact details, please ask Admin. You can change your password from the Settings menu.

Gym-Pro | Created By: Nenshi Pal

- Announcement

← → ↻ localhost:8080/gym-management/trainer/announcements.php 🔍 🖨 ⭐ 📱 🌙

GYM-PRO

TRAINER PANEL

- Dashboard
- My Batches
- My Members
- Today's Schedule
- Workout Plans
- Mark Attendance
- Announcements**
- My Profile
- Change Password
- View Public Website
- Logout

Announcements

Post New Announcement

Post

Welcome to Gym-Pro

You can now join more than one batch — as long as their timings do not overlap. Check the Batches page to see what is available.

Posted by Admin on 2026-09-17 12:14:44

Gym-Pro | Created By: Nenshi Pal

- Change Password

← → C localhost:8080/gym-management/trainer/settings/change-password.php

GYM-PRO

TRAINER PANEL

- Dashboard
- My Batches
- My Members
- Today's Schedule
- Workout Plans
- Mark Attendance
- Announcements
- My Profile
- Change Password**
- View Public Website
- Logout

Change Password

Rahul Sharma

Current Password

New Password

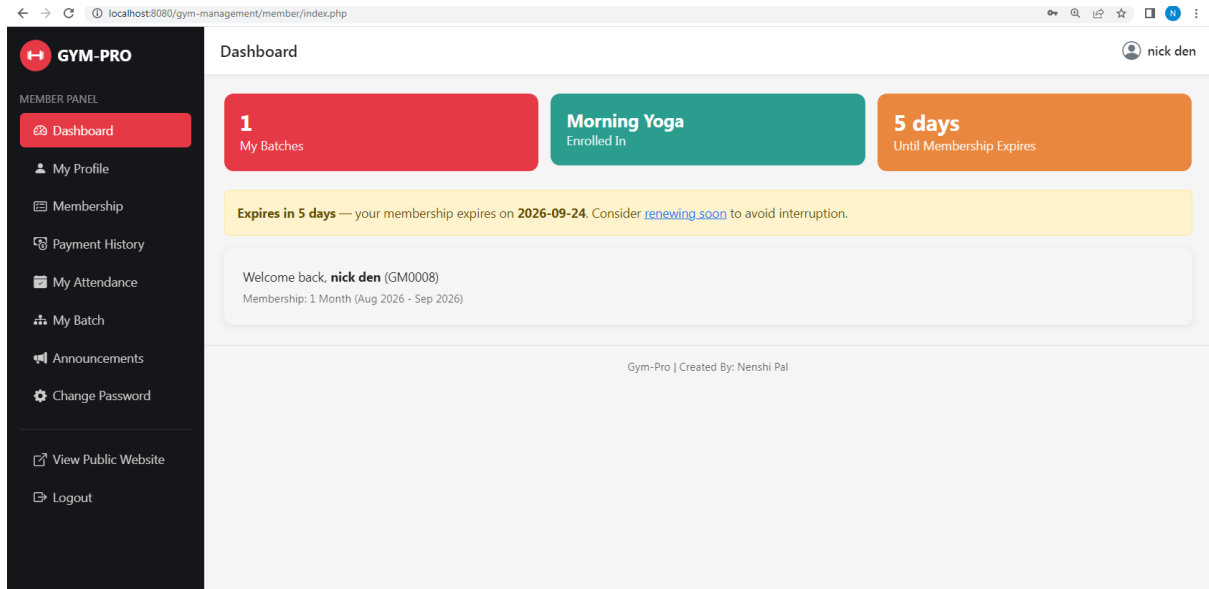
Confirm New Password

Update Password

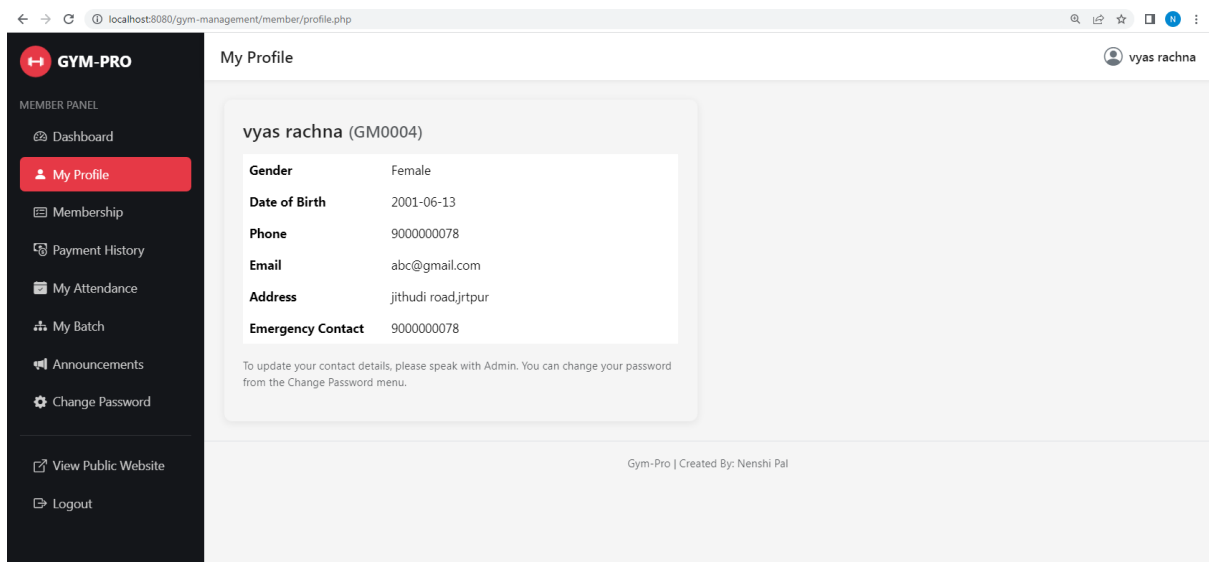
Gym-Pro | Created By: Nenshi Pal

4. Member

- Dashboard



- My Profile



- My Membership

Membership

vyas rachna

Your Batches

Morning Yoga (06:00-07:00)
Morning Cardio (07:15-08:15)

Duration 1 Month (Sep 2026 - Oct 2026)

Start Date 2026-09-17

Expiry Date 2026-10-17

Status Active

[Request Renewal / Change Batches](#)

Gym-Pro | Created By: Nenshi Pal

- Payment History

Payment History

vyas rachna

Rs. 1,500.00
Total Due (current batches)

Rs. 1,500.00
Total Paid

Rs. 0.00
Pending Balance

Receipt #	Amount	Date	Method	Notes
RC0004	Rs. 1,500.00	2026-09-01	UPI	Verified online payment — Transaction ID: 4545454545

Gym-Pro | Created By: Nenshi Pal

• My Attendance

The screenshot shows the 'My Attendance' page of the GYM-PRO application. The left sidebar contains a 'MEMBER PANEL' with links to Dashboard, My Profile, Membership, Payment History, My Attendance (highlighted), My Batch, Announcements, Change Password, View Public Website, and Logout. The main content area displays the user's attendance for 'Morning Yoga (1/1 days)' with a 100% completion status. Below this, a table titled 'Recent Attendance' shows a single record for '2026-09-19' with the status 'Present'. The footer of the page reads 'Gym-Pro | Created By: Nenshi Pal'.

Date	Batch	Status
2026-09-19	Morning Yoga	Present

• My Batch

The screenshot shows the 'My Batches' page of the GYM-PRO application. The left sidebar is identical to the previous page, with 'My Batch' highlighted. The main content area displays details for the 'Morning Yoga' batch, including the time '06:00 - 07:00'. It lists the 'Trainers' as 'Priya Nair (Yoga & Flexibility)' and provides a 'Weekly Workout Plan' for 'Tuesday' focusing on 'Flexibility & Breathing' with a duration of 'Sun salutations, standing poses, pranayama 15 min'. The footer of the page reads 'Gym-Pro | Created By: Nenshi Pal'.

- Pending Payment

The screenshot displays the 'Complete Your Payment' interface. On the left is a dark sidebar with the 'GYM-PRO' logo and a 'MEMBER PANEL' containing links for 'Membership Expired', 'Renewal Status', 'View Public Website', and 'Logout'. The main content area is titled 'Complete Your Payment' and features a dark header with a lock icon and the text 'Complete Your Payment to Activate'. Below this, a table lists the order details: Morning Yoga (Rs. 800.00 / month), Evening Strength (Rs. 1,000.00 / month), Total Amount (Rs. 1,800.00), and Paid Amount (Rs. 0.00). A red 'Remaining Amount' of Rs. 1,800.00 is shown. A yellow warning box states: 'Your last submission was rejected. Please double-check your payment and submit again below.' Below this is a QR code with the text 'Scan with any UPI app to pay for gympro@upi' and 'Amount to pay: Rs. 1,800.00'. A 'SUBMIT TRANSACTION' section includes a 'Payment Method' dropdown set to 'UPI / QR Payment', an 'Amount Paid' field with 'Rs. 1,800.00', a 'Cancellation Amount' field with 'Rs. 1,800.00', and a 'Transaction ID / UTR' field with a sample ID '123456789012'. A red 'Submit Payment' button is at the bottom. A blue note at the very bottom states: 'This system does not process the payment itself -- pay via the QR/URI above, then submit your transaction ID. Admin verifies it against the gym's account and your account unlocks automatically.'

- Renewal – Request

The screenshot shows the 'Renewal Request' page. The sidebar is identical to the previous page. The main content area has a header 'Renewal Request' and a user profile 'isha gondaliya'. The form includes a 'Duration (months)' dropdown set to '1'. Below this, it says 'Select Batches (max 3) — non-overlapping only'. There are two columns of batch options: 'MORNING' and 'EVENING'. The Morning column lists 'Morning Yoga (06:00-07:00) Rs. 800.00 / month', 'Morning Strength (06:00-07:00) Rs. 1,000.00 / month', and 'Morning Cardio (07:15-08:15) Rs. 700.00 / month'. The Evening column lists 'Evening Zumba (17:00-18:00) Rs. 900.00 / month' and 'Evening Strength (18:15-19:15) Rs. 1,000.00 / month'. Each option has an unchecked checkbox. A red 'Submit Request' button is at the bottom of the form. The footer of the page reads 'Gym-Pro | Created By: Nenshi Pal'.

6. Software Testing

6.1 manual testing

Black Box Testing is performed by providing inputs to the system and verifying whether the resulting behaviour matches the expected functionality. The internal implementation of the system is not considered during this testing.

For GYM PRO – Gym Management System, Black Box Testing is presented in two parts below: Input Validation Testing, which checks how each form reacts to valid, blank, invalid and boundary input; and Functional Testing, which checks that each feature performs its intended operation correctly.

Result: The tested Black Box functions and validation scenarios produced the expected results.

Black Box Testing – Input Validation Testing

Every field of every form was tested using four input types:

- Input Type 1: Valid input
- Input Type 2: Mandatory field left blank
- Input Type 3: Invalid input
- Input Type 4: Optional field / default value / boundary condition

Modules covered:

- Registration
- Login
- Add Batch
- Add Member (Admin)
- Assign Trainer to Batch
- Approve / Reject Renewal Request
- Log Payment
- Online Payment Submission
- System Settings

1. Registration

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Full Name	Yes	No	Yes	Yes
Phone	Yes	Yes	Yes	No
Email	Yes	Yes	No	Yes
Date of Birth	Yes	No	Yes	Yes
Username	Yes	No	Yes	Yes
Password	Yes	No	Yes	Yes
Batch Selection	Yes	Yes	Yes	Yes
Result	Pass	Fail	Fail	Pass

Criteria: Full Name, Phone, Email, Date of Birth, Username, Password and at least one Batch are mandatory. Date of Birth must satisfy the minimum age of 18 years (boundary condition), and the selected Batch(es) must not exceed the maximum allowed or overlap in timing.

2. Login (Admin / Trainer / Member)

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Username	Yes	Yes	No	No
Password	Yes	No	Yes	No
Result	Pass	Fail	Fail	Fail

Criteria: Username and Password are mandatory.

3. Add Batch

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Batch Name	Yes	No	Yes	Yes
Start Time	Yes	Yes	Yes	No
End Time	Yes	Yes	Yes	No
Capacity	Yes	Yes	No	Yes
Monthly Fee	Yes	Yes	No	Yes
Result	Pass	Fail	Fail	Pass

Criteria: Batch Name, Start Time and End Time are mandatory, and End Time must be later than Start Time (invalid condition). Capacity defaults to 20 and Monthly Fee defaults to 0 when left blank (boundary/default condition).

4. Add Member (Admin)

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Full Name	Yes	No	Yes	Yes
Phone	Yes	Yes	Yes	No
Email	Yes	Yes	No	Yes
Duration (months)	Yes	Yes	No	Yes
Batch Selection	Yes	Yes	Yes	Yes
Result	Pass	Fail	Fail	Pass

Criteria: Full Name, Phone, Email and at least one Batch are mandatory. Duration defaults to 1 month if left blank; the member created is still placed in Pending Payment status until a payment is logged.

5. Assign Trainer to Batch

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Batch	Yes	No	Yes	Yes
Trainer	Yes	Yes	No	Yes
Result	Pass	Fail	Fail	Pass

Criteria: Batch and Trainer are mandatory; the same Trainer cannot be assigned twice to the same Batch (unique constraint).

6. Approve / Reject Renewal Request

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Renewal Request	Yes	No	Yes	Yes
Decision (Approve/Reject)	Yes	Yes	No	Yes
Requested Duration	Yes	Yes	No	Yes
Result	Pass	Fail	Fail	Pass

Criteria: Renewal Request and Decision are mandatory; Requested Duration has a boundary condition of 1 to 12 months.

7. Log Payment

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Member	Yes	No	Yes	Yes
Amount	Yes	Yes	Yes	No
Payment Date	Yes	Yes	Yes	No
Payment Method	Yes	Yes	No	Yes
Result	Pass	Fail	Fail	Pass

Criteria: Member, Amount and Payment Date are mandatory; Payment Method defaults to Cash when left blank.

8. Online Payment Submission

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Transaction ID	Yes	Yes	Yes	No
Payment Date	Yes	Yes	Yes	No
Result	Pass	Fail	Fail	Fail

Criteria: Transaction ID and Payment Date are mandatory. Amount is system-calculated from the member's due fee and cannot be edited by the member (default value).

9. System Settings

Field Name	Input Type 1	Input Type 2	Input Type 3	Input Type 4
Setting Key	Yes	No	Yes	Yes
Setting Value	Yes	Yes	No	Yes
Result	Pass	Fail	Fail	Pass

Criteria: Setting Key is mandatory and unique; Setting Value is mandatory and cannot be NULL according to the database design.

Black Box Testing – Functional Testing

Test Type meaning:

Test Type	Description
Test Type 1	Normal / successful operation
Test Type 2	Missing / incomplete operation
Test Type 3	Invalid / incorrect operation
Test Type 4	Alternate / exceptional operation

Modules covered:

- Registration
- Login
- Batch Management
- Member Management
- Trainer Assignment
- Renewal Approval
- Attendance Management
- Payment Submission & Verification
- Search and Filter
- Logout

1. Registration

Function / Operation	Test Type 1	Test Type 2	Test Type 3	Test Type 4
Enter valid registration details	Yes	No	Yes	Yes
Create member record	Yes	No	No	Yes
Display validation message	No	Yes	Yes	No
Result	Pass	Pass	Pass	Pass

Criteria: The system should create a Pending Payment member record when valid details are entered, and display appropriate validation messages for missing or invalid details.

2. Login

Function / Operation	Test Type 1	Test Type 2	Test Type 3	Test Type 4
Enter username	Yes	Yes	No	No
Enter password	Yes	No	Yes	No
Authenticate user	Yes	No	No	No
Display error message	No	Yes	Yes	Yes
Result	Pass	Pass	Pass	Pass

Criteria: The system should authenticate valid credentials for the correct panel (Admin / Trainer / Member) and display an appropriate error message when credentials are missing or incorrect.

3. Batch Management

Function / Operation	Test Type 1	Test Type 2	Test Type 3	Test Type 4
Add batch	Yes	No	No	No
View batch	Yes	Yes	No	No
Update batch	No	No	Yes	No
Delete batch	No	No	No	Yes
Result	Pass	Pass	Pass	Pass

Criteria: Admin should be able to add, view, update and delete batches successfully; deleting a batch cascades its trainer assignments and schedules.

4. Member Management

Function / Operation	Test Type 1	Test Type 2	Test Type 3	Test Type 4
Add member	Yes	No	No	No
View member	Yes	Yes	No	No
Update member	No	No	Yes	No
Delete member	No	No	No	Yes
Result	Pass	Pass	Pass	Pass

Criteria: Admin should be able to add, view, update and delete member records; each new or renewed member passes through the Pending Payment gate.

5. Trainer Assignment

Function / Operation	Test Type 1	Test Type 2	Test Type 3	Test Type 4
Select batch	Yes	No	Yes	Yes
Select trainer	Yes	Yes	No	Yes
Assign trainer	Yes	No	No	Yes
Display assignment	Yes	No	No	Yes
Result	Pass	Pass	Pass	Pass

Criteria: Admin should be able to select a batch and assign an available trainer to it, with the assignment displayed correctly and duplicate assignments prevented.

6. Renewal Approval

Function / Operation	Test Type 1	Test Type 2	Test Type 3	Test Type 4
Pending → Approved	Yes	No	No	No
Pending → Rejected	No	Yes	No	No
Reject invalid duration	No	No	Yes	No
Display updated status / expiry	No	No	No	Yes
Result	Pass	Pass	Pass	Pass

Criteria: Admin should be able to approve or reject a renewal request, with the member's expiry date extended automatically only on approval.

7. Attendance Management

Function / Operation	Test Type 1	Test Type 2	Test Type 3	Test Type 4
Mark attendance for a batch	Yes	No	No	No
Leave a member unmarked	No	Yes	No	No
Mark attendance twice, same day	No	No	Yes	No
View attendance history	No	No	No	Yes
Result	Pass	Pass	Pass	Pass

Criteria: Trainer should be able to mark attendance for their assigned batch once per member per day; both trainer and member should be able to view attendance history.

8. Payment Submission & Verification

Function / Operation	Test Type 1	Test Type 2	Test Type 3	Test Type 4
Submit online payment	Yes	No	Yes	Yes
Verify payment submission	Yes	No	No	Yes
Reject payment submission	No	No	Yes	Yes
View submission status	Yes	Yes	No	Yes
Result	Pass	Pass	Pass	Pass

Criteria: Member should be able to submit an online payment claim, and Admin should be able to verify or reject it; the member's status becomes Active only after verification or an in-person payment is logged.

9. Search and Filter

Function / Operation	Test Type 1	Test Type 2	Test Type 3	Test Type 4
Search by member code / name	Yes	No	No	No
Filter by status	No	Yes	No	No
Filter by batch	No	No	Yes	No
Filter by payment status	No	No	No	Yes
Result	Pass	Pass	Pass	Pass

Criteria: The system should correctly search members by name or code and filter records by status, batch and payment status.

10. Logout

Function / Operation	Test Type 1	Test Type 2
User logged in	Yes	No
Click logout	Yes	No
Session terminated	Yes	Yes
Result	Pass	Pass

Criteria: A logged-in user (Admin, Trainer or Member) should be able to log out successfully, and the active session should be terminated.

6.2 Database Testing

Database Testing verifies the data integrity, constraints, relationships and default values defined in the gym_pro database.

The schema defines mandatory fields and unique constraints for members, including member_code, phone, email and username; gender, duration_months and status have default values. The payments and payment_submissions tables define mandatory member, amount and date fields, a unique receipt_number, and foreign-key relationships back to members and admins.

The foreign-key rules in the schema explicitly use CASCADE (e.g. deleting a member removes their payments, attendance and batch enrolments) and SET NULL (e.g. deleting the admin who verified a payment leaves the payment record in place with verified_by set to NULL), so these are genuine database tests rather than generic examples.

1. Members Table

Test Case ID	Test Condition / Action	Expected Result	Actual Result	Status
DB-MEM-01	Insert member without Full Name	Record should be rejected	Record was rejected	Pass
DB-MEM-02	Insert member with duplicate Phone	Duplicate should be rejected	Duplicate was rejected	Pass
DB-MEM-03	Insert member with duplicate Email	Duplicate should be rejected	Duplicate was rejected	Pass
DB-MEM-04	Insert member without Duration	Default of 1 month should be applied	Default of 1 month was applied	Pass
DB-MEM-05	Insert member without Status	Default 'Active' should be applied	Default 'Active' was applied	Pass

2. Admins Table

Test Case ID	Test Condition / Action	Expected Result	Actual Result	Status
DB-ADM-01	Insert admin without Username	Record should be rejected	Record was rejected	Pass
DB-ADM-02	Insert admin with duplicate Username	Duplicate should be rejected	Duplicate was rejected	Pass
DB-ADM-03	Insert admin without Password	Record should be rejected	Record was rejected	Pass

3. Trainers Table

Test Case ID	Test Condition / Action	Expected Result	Actual Result	Status
DB-TRN-01	Insert trainer without Full Name	Record should be rejected	Record was rejected	Pass
DB-TRN-02	Insert trainer with duplicate Phone	Duplicate should be rejected	Duplicate was rejected	Pass
DB-TRN-03	Insert trainer without Status	Default 'Active' should be applied	Default 'Active' was applied	Pass

4. Batches Table

Test Case ID	Test Condition / Action	Expected Result	Actual Result	Status
DB-BAT-01	Insert batch without Batch Name	Record should be rejected	Record was rejected	Pass
DB-BAT-02	Insert batch without Capacity	Default 20 should be applied	Default 20 was applied	Pass
DB-BAT-03	Insert batch without Monthly Fee	Default 0 should be applied	Default 0 was applied	Pass

5. Payments Table

Test Case ID	Test Condition / Action	Expected Result	Actual Result	Status
DB-PAY-01	Insert payment without Member	Record should be rejected	Record was rejected	Pass
DB-PAY-02	Insert payment with duplicate Receipt Number	Duplicate should be rejected	Duplicate was rejected	Pass
DB-PAY-03	Insert payment without Payment Method	Default 'Cash' should be applied	Default 'Cash' was applied	Pass
DB-PAY-04	Insert payment with nonexistent Member ID	Foreign-key constraint should reject record	Record was rejected	Pass

Deleting a Member cascades to their Payments (ON DELETE CASCADE); deleting the Admin who recorded a payment also cascades on this table.

6. Payment Submissions Table

Test Case ID	Test Condition / Action	Expected Result	Actual Result	Status
DB-PSUB-01	Insert submission without Transaction ID	Record should be rejected	Record was rejected	Pass
DB-PSUB-02	Insert submission without Status	Default 'Pending Verification' should be applied	Default 'Pending Verification' was applied	Pass
DB-PSUB-03	Insert submission with nonexistent Member ID	Foreign-key constraint should reject record	Record was rejected	Pass

Deleting the Admin who verified a submission sets verified_by to NULL (ON DELETE SET NULL); the submission record itself is kept.

7. Renewal Requests Table

Test Case ID	Test Condition / Action	Expected Result	Actual Result	Status
DB-REN-01	Insert renewal request without Member	Record should be rejected	Record was rejected	Pass
DB-REN-02	Insert renewal request without Status	Default 'Pending' should be applied	Default 'Pending' was applied	Pass
DB-REN-03	Insert renewal request with nonexistent Member ID	Foreign-key constraint should reject record	Record was rejected	Pass

8. Settings Table

Test Case ID	Test Condition / Action	Expected Result	Actual Result	Status
DB-SET-01	Insert setting without Setting Key	Record should be rejected	Record was rejected	Pass
DB-SET-02	Insert setting with duplicate Setting Key	Duplicate should be rejected	Duplicate was rejected	Pass
DB-SET-03	Insert setting with NULL Setting Value	Record should be rejected	Record was rejected	Pass
DB-SET-04	Insert valid setting	Setting should be stored	Setting was stored successfully	Pass

The schema defines setting_key as NOT NULL and UNIQUE, and setting_value as NOT NULL — unlike a generic key-value design, no setting in GYM PRO may be stored with a blank value.

6.3 Test Summary

Testing Type	Total Test Cases	Passed	Failed
Black Box Testing – Input Validation	36	36	0
Black Box Testing – Functional	38	38	0
Database Testing	28	28	0
Total	102	102	0

Testing Conclusion

GYM PRO – Gym Management System was tested using Black Box Testing and Database Testing. The testing covered input validation across every form, the core functional operations of the Admin, Trainer and Member panels, and database constraints, default values, unique fields and referential integrity, including the payment-gate rule that keeps a member in Pending Payment until a payment is logged or verified. The documented test cases produced the expected results.

7. Enhancements

7.1 Advantages

- All database queries use prepared statements (mysqli), which protects the system against SQL-injection attacks.
- Passwords are stored using PHP's password_hash()/password_verify() functions and are never kept in plain text.
- Each panel — Admin, Trainer and Member — has its own independent session-based login; an unauthorized visitor is redirected back to that panel's login page.
- The core business rule — that a member cannot join two batches whose timings overlap — is enforced on the server side, so it cannot be bypassed even if JavaScript is disabled in the browser.
- The payment gate applies with no exceptions: every path that creates or renews a member, including the Admin's own Add Member and Renew Member actions, lands in Pending Payment first, so no membership can become Active without a logged or verified payment.
- An online payment is never trusted on its own — the member only submits a transaction ID, and nothing is treated as a real payment until the Admin verifies it, which keeps a human check in the loop.
- A trainer can view and manage only the batches actually assigned to them, not the entire gym's data.
- The system is built to run on both modern PHP (7.x / 8.x) and older PHP 5.4 installations, which is useful in college lab environments.

7.2 Limitation

- There is still no real payment gateway integration; online payment is a submit-and-verify claim against the gym's own UPI ID, not an automated checkout.
- Photo upload for members and trainers is not yet implemented, although the database already has a column reserved for it.
- No automatic email or SMS notification is sent when a registration or renewal request is approved or rejected, or when a payment is verified — the member has to log in to check the status.
- Long lists (members, trainers, requests) do not yet have pagination, which may slow down as the number of records grows very large.
- The Contact Us form on the public site only shows a confirmation message; enquiries are not yet stored in the database for the Admin to review later.

7.3 Future Scope

- Add real photo upload for members and trainers using PHP's move_uploaded_file() function.
- Integrate an actual payment gateway (for example Razorpay) so an online payment is confirmed automatically instead of waiting for Admin verification.
- Send an automatic email or SMS notification when a registration/renewal request is approved or rejected, or when a payment submission is verified or rejected.
- Add a contact_messages table and an Admin inbox so enquiries from the Contact Us page can be tracked and answered.
- Add pagination and search/filter options to all the long list pages.
- Package the member-facing pages as a Progressive Web App (PWA) so members can check their schedule, attendance and payment status conveniently on mobile.

8. Bibliography

Websites:

<https://www.php.net>

<https://www.w3schools.com>

<https://getbootstrap.com>

<https://dev.mysql.com/doc/>

<https://www.chartjs.org>

<https://davidshimjs.github.io/qrcodejs/>

Books:

- PHP and MySQL Web Development – Luke Welling & Laura Thomson
- Learning PHP, MySQL & JavaScript – Robin Nixon